

Review

Ivaturi Siva Ramakoti, Achyut Kumar Panda and Narayan Gouda*

A brief review on polymer nanocomposites: current trends and prospects

<https://doi.org/10.1515/polyeng-2023-0103>

Received April 20, 2023; accepted July 18, 2023;

published online August 10, 2023

Abstract: Nanoparticles (NPs) are being incorporated into pristine polymeric matrices as well as recycled waste polymers to prepare new materials with enhanced properties called polymer nanocomposites (PNCs). There has been an accelerating growth in the field of polymer nanocomposites in the last two decades owing to their tunable and enhanced optical, thermal, electrical, magnetic, and mechanical properties, which make them viable and well-suited for various applications. The addition of nanoparticles in the form of nanosheets, nanotubes, nanospheres, quantum dots, etc., to the polymer matrices reduces the drawbacks of polymers like low mechanical strength, weaker chemical resistivity, etc. due to the large surface area to volume ratio of NPs. Moreover, the interaction of nanoparticles with the polymer chain lowers the permeability of gases and increases the water resistance of the polymer. Due to their remarkable capabilities, polymer nanocomposites have attracted significant theoretical interest as well as a large number of practical applications in a variety of scientific and technological domains. This article provides an overview of the various classes of nanoparticles and different fabrication methods used to produce polymer nanocomposites. Additionally, an effort is made to comprehend the properties and applications of polymer nanocomposites in different areas like wastewater treatment, dye removal, heavy metal and gas sensing, optoelectronics, etc.

Keywords: dye removal; gas sensing; nanoparticles; polymer nanocomposites; water treatment

1 Introduction

The exponential increase in research and publications over the last 10 years demonstrates the tremendous momentum that nanoparticles and polymer nanocomposites have gained in a wide range of disciplines. Nanoparticles are materials with one dimension in the nanoscale regime, more precisely within 100 nm. The relationship between the properties, size, shape, and surface structure of nanoparticles is especially intriguing. They do, however, have a propensity to agglomerate due to their ultrafine size, especially if they are not well stabilised both during synthesis and application. This is due to their high surface reactivity and energy, both of which are reduced by the agglomeration process. The surface-to-volume ratio of nanoparticles, also known as their aspect ratio, has a significant impact on their activity, particularly how they interact with other materials. Polymeric nanocomposites are created by the uniform dispersal of NPs in pristine polymer matrices where at least one phase is in the nanoscale dimension. The incorporation of NPs into the polymer matrix significantly improves the majority of the desired properties of pristine polymers, which include biodegradability, thermal, flame retardancy, and mechanical properties, among others. The incorporation of NPs may also generate a set of novel properties due to the nature and size of the nanoparticles used. The surface chemistry and physicochemical intricacies that are altered and tailored when nanoparticles are added to polymer matrices are intimately related to the performance of the resulting polymeric materials. The shape, size, surface properties, and aspect ratio of nanoparticles are critical factors in adjusting these interactions and also in the properties of the resulting systems. Polymer nanocomposite performance outperforms that of traditional filled and composite polymeric systems, putting it in a class of its own. The term “nanocomposite” became widely accepted in the scientific community during the year 1992 [1]. Because of the significant interfacial interactions at the interfaces of nanoparticles and polymer matrices, polymer nanocomposites (PNCs) differ fundamentally from traditional composites. As a result of the active

*Corresponding author: **Narayan Gouda**, Department of Chemistry, Centurion University of Technology and Management, Paralakhemundi, Odisha PIN-761211, India, E-mail: gouda24narayan@gmail.com. <https://orcid.org/0000-0001-6245-3425>

Ivaturi Siva Ramakoti, Department of Chemistry, Centurion University of Technology and Management, Paralakhemundi, Odisha PIN-761211, India

Achyut Kumar Panda, Department of Chemistry, Veer Surendra Sai University of Technology, Burla, Sambalpur, Odisha, India

combination of NPs and polymers, polymer nanocomposites now have a wide range of application-oriented utilities. Nanoparticles and polymer nanocomposites play a pivotal role in the current state of materials development and have a wide range of applications that include packaging, catalysis, sensing, energy storage systems, optoelectronics, flame retardant materials, corrosion-resistant surface coatings, stimuli-responsive materials, drug delivery systems, and so on [2]. Nanoparticles are substantially more active than their bulk counterparts, as most chemical interactions occur on the surface of materials. In this context, the catalytic activity of nanoparticles can be used as an example. One of the most surprising aspects of nanoparticles is their size dependence. Depending on its size, the same nanomaterial can have multiple colors. Furthermore, the shape and aspect ratio of nanomaterials has a significant impact on how they interact with other materials, such as polymers. Incorporating nanoparticles in very small amounts (0.1–5 wt%) into other materials, such as polymers, enhances many of their characteristics while also making processing considerably simpler than with traditional systems and maintaining their modest weight. When appropriate polymer nanocomposites are formed from the respective virgin polymers, attributes such as mechanical strength, toughness, scratch resistance, thermal stability, chemical resistance, gas barrier, and so on are frequently greatly improved. There are numerous examples in the literature where transparency and lightweight properties are preserved while other properties such as flame retardancy, biodegradability, and so on are improved. Depending on the nanoparticles used, polymer nanocomposites may also significantly improve biocompatibility and produce novel and intriguing features such as magnetic, electric, catalytic, optical, and so on without significantly increasing overall cost. As a result, when compared to traditional fillers or composites, nanoparticles and associated polymeric nanocomposites have a much lower cost-to-performance ratio. The incredibly large interfacial area and nanoscopic size of the fillers distinguish polymer nanocomposites from ordinary composites. Because of the dominance of interfacial areas caused by nanoscale phase dimensions, some unusual mechanical and physical properties have been generated that are not present in traditional polymer composites. Because of the extremely large interfacial area between polymers and nanofillers, the functionalization of nanofillers is used to change interfacial states to improve the mechanical and physical properties of polymer nanocomposites [3].

1.1 Trend analysis of publication on polymer nanocomposite technology

Over the last decade, publications on nanoparticles and polymer nanocomposites have exponentially increased, demonstrating the immense momentum they have received in a variety of sectors. This can be visualized in Figure 1. This growth is primarily because there are various sorts of commercially available nanoparticles that can be integrated into the polymer network to frame polymer nanocomposites. Contingent upon the nanoparticles utilised, polymer nanocomposites may likewise essentially further develop biocompatibility and make new as well as intriguing properties, including catalytic, optical, electrical, magnetic, and so forth, without fundamentally expanding the general expenses. Since nanoparticles and their polymeric nanocomposites frequently have execution costs that are substantially lower than those of conventionally filled or composite frameworks, these materials have a very high potential for commercial use. The main goal of PNC's research is to investigate various strategies for significantly improving characteristics compared to those of conventional matrix polymers while employing only a tiny amount of nanoparticles. More novel physical and chemical features than those found in conventionally filled polymeric materials are anticipated from these nanoparticles. From decades of PNC research, it is well recognised that some nanoparticles only enhance a single type of property, such as mechanical, thermal, flammability, or electrical, whereas a small number of nanoparticles can help enhance

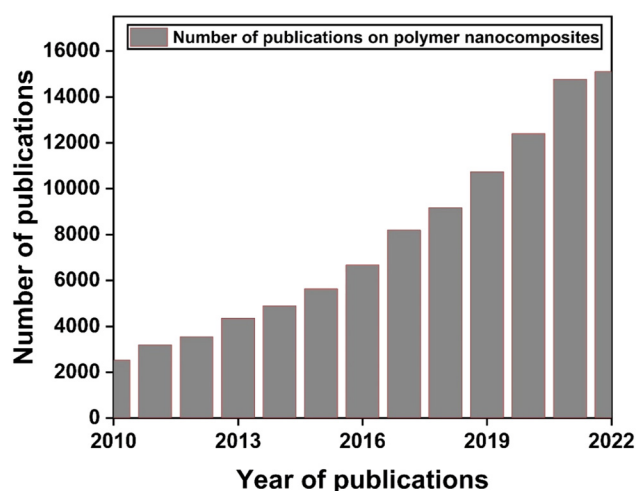


Figure 1: Plot showing the work published on PNCs (obtained from Science Direct).

multifunctional capabilities. In comparison to conventionally filled polymers, they provide a radical alternative. In contrast to conventional polymer systems, where reinforcement is on the order of microns, PNCs are exemplified by distinct constituents on the order of a few nanometers [4, 5].

2 Types and sources of polymer used

Almost all types of polymers, including plastics, rubbers, and fibers, are used as matrices to generate polymer nanocomposites. The matrices for various polymer nanocomposites can also be made of high- and low-molecular-weight elastomers and resins. Very often, the matrices used in PNCs are pre-made polymers as well as *in situ*-produced (in the presence of nanoparticles) polymers. In the last few years, recycled waste polymers derived from waste plastics or single-use plastics have also been used as a matrix for polymer nanocomposites. Recycled waste polymers have been used by many research teams as fillers and additives, or they have been combined with nanofillers to create polymer matrix nanocomposites (PNCs) with improved qualities while reducing the negative effects of these plastic wastes. Recent research and studies have shown that adding a small amount of nanoparticles to the polymers has resulted in the formation of polymer nanocomposites with better mechanical, thermal, optical, barrier, and adsorbing properties than both the virgin polymer and the single-use plastic products. Some of the work on PNCs based on recycled polymers, along with their properties and applications, is given in Table 1 [6]. In addition to being a continuous phase, the matrix of a nanocomposite stabilises and interacts actively with the nanomaterial, and the final properties of the PNCs depend upon many factors associated with the polymer matrix material like crystallinity, polymer morphology, molecular weight, and type of the polymer. Apart from the factors mentioned above, the polymer synthesis methods such as melt mixing, solution casting, *in-situ* polymerization, and emulsion polymerization also play an important role in the property determination of the final PNCs. Various homopolymers (polymers with one type of monomer) like polyethylene (PE) [7], polypropylene (PP) [8], polystyrene (PS) [9], polyvinyl chloride (PVC) [10], polymethyl methacrylate (PMMA) [11], polyacrylonitrile (PAN) [12], polybutadiene [13], poly tetrafluoroethylene (PTFE) [14], poly vinylidene fluoride (PVDF) [15], polyisoprene [16], polychloroprene [17], and copolymers (polymers with more

Table 1: Properties and applications of PNC from recycled polymers.

PNC	Properties	Application	References
Recycled LDPE (low density polyethylene)/corn husk fibre	Increase in elastic modulus and tensile strength	Packaging material	[26]
Recycled PET (polyethylene terephthalate)/MWCNT (multi-walled carbon nanotube)	Higher thermal stability	Wastewater treatment	[27]
Recycled PET/nano-ZnO	UV-stability	Packaging	[28]
Recycled polystyrene/TiO ₂	Photo catalytic	Photocatalyst	[29]
Recycled PET/nanoclay	Viscosity and mechanical properties	Extrusion and injection moulding applications	[30]
Recycled high-density polyethylene (rHDPE)/graphene/MWCNT	Thermal degradation temperature, improved stiffness	Thermal storage systems, automobile, packaging	[31]

than one type of monomer) like ethylene-propylene copolymer [18, 19], poly(acrylonitrile-butadiene-styrene-butadiene-styrene-butadiene) [20], poly(styrene-butadiene) [21], poly(vinylidene fluoride-hexafluoropropylene-te) [22], poly(acrylonitrile-butadiene) [23], etc. It is important to mention that the use of a copolymer like poly(styrene-butadiene) [24] or poly(acrylonitrile-butadiene) [25] as the polymer matrix in the PNCs preparation gives rise to not only flexible polymers but also numerous better qualities compared to the individual homopolymers due to the internal plasticization effect.

3 Nanofillers used, their classification, and synthesis

The preparation of diverse polymer nanocomposites uses nearly all kinds of nanoparticles, and the type of nanomaterial needed to provide the desired impact would depend on the application, according to the study. The most widely used nanoparticles are metals (Au, Ag), metal oxides (alumina, ZnO, TiO₂, etc.), nanoclays, and carbon-based nanoparticles like graphene, carbon nanotubes, carbon dots, etc. Consequently, a quick rundown of a few significant such nanoparticles is briefly explained below.

3.1 Metal-based nanoparticles

Metal-based nanoparticles like gold (Au), silver (Ag), iron (Fe), and copper (Cu) are used as dispersed phases in the preparation of polymer nanocomposites. Metal-based nanoparticles are generally used or dispersed in polymer matrices to impart and improve certain properties, like catalytic, optical, antimicrobial, etc. For example, metal nanoparticles like Ag and Au are well known to have optical characteristics due to their localised surface plasmon effects. These plasmonic nanoparticles can react to their environment or even affect optical processes such as fluorescence, absorption, and molecular activity at their surface, including Raman scattering. When these plasmonic metal nanoparticles are dispersed in the polymer matrix, the resultant plasmonic polymer nanocomposites show diverse phenomena that are further utilised in sensors for the detection of chemicals in biological sensors and also as efficient light-harvesting materials in solar cells [32]. Ag nanoparticles have excellent antimicrobial properties. Ismayil et al. [33] used the non-solvent-induced phase inversion technique to prepare silver nanoparticle-doped polyaniline-PVC nanocomposite films. The AgNP-doped nanocomposites displayed good photodegradation and superior antimicrobial effectiveness. Similarly, Awad et al. [34] prepared Ag-PMMA nanocomposites, and their research showed that the thermal stability of the AgNP/PMMA nanocomposite is greater than that of the PMMA polymer. The nanocomposite demonstrated strong antibacterial activity against bacteria.

3.2 Metal oxide-based nanoparticles (MONPs)

Metal oxides such as aluminum oxide (Al_2O_3), iron oxide (Fe_2O_3 or Fe_3O_4), copper oxide (CuO or Cu_2O), zinc oxide (ZnO), titanium dioxide (TiO_2), silicon dioxide or silica (SiO_2), and gallium oxide (Ga_2O_3), are the few examples of MONPs that are generally utilised in the synthesis of PNCs. These MONPs have rendered excellent mechanical, optical, catalytic, and electrical which leads to their wide applicability in fuel cells, advanced ceramics, solar cells, pyroelectric, supercapacitors, catalysts, and anti-corrosion coatings, etc. The characteristics of PNCs can be enhanced by the addition of metal oxide nanoparticles [35]. The fabrication of nanocomposites with various polymer matrices has been shown to improve a variety of properties such as toughness, thermal stability, glass transition temperature (T_g), tensile strength, and so on. Maharramov et al. prepared polymer

nanocomposites based on isotactic polypropylene and Fe_3O_4 nanoparticles and studied the structure and dielectric properties of these prepared composites. Their studies revealed that with the addition of Fe_3O_4 nanoparticles into the polypropylene matrix, the dielectric permittivity (ϵ) of the composite increases with increasing nanoparticle concentration, and with increasing frequency, the value of " ϵ " remains unchanged [36]. Using the simple electrospinning technique, recycled polystyrene foam with various proportions of titanium dioxide nanoparticles (TiO_2 NPs) and aluminum microparticles was spun into superhydrophobic nanocomposite fibres [37].

Özge Lalegül-Ülker et al. have developed Silica coated iron oxide/polyaniline (PANI) polymer nanocomposites and their research revealed that silica acted as a ceramic coating component that improved the *in vitro* biocompatibility of the nanocomposites as well as their physical and chemical characteristics. On the other hand, the nanocomposite structures were given electrical conductivity and magnetic susceptibility by polyaniline and Si- Fe_3O_4 NPs, respectively so these nano-composites may be used in a range of biomedical applications, depending on their electrical-magnetic properties and biocompatibility [38].

3.3 Nano clays

Nano clays are layered silicates that have a very fine size which is below 100 nm. In accordance with their shape and chemical composition nano clays come in many different varieties, including kaolinite, montmorillonite, bentonite, hectorite, and halloysite [39, 40]. Montmorillonite, bentonite, hectorite, saponite, smectite, phyllosilicate, and kaolin are typical nano clay minerals. The most popular nanofillers among them are kaolin, montmorillonite [41]. Incorporating nano clay into the polymer matrices improves properties such as heat resistance stiffness and reduces the permeability of gases (oxygen or water vapor) [42]. Table 2 shows the various properties that are enhanced with the addition of nano clays into the polymer matrix. These nano clays have sheet-like geometry. The kaolinite group of nano clay has the general formula $\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})$ and the montmorillonite group with the general formula $(\text{Ca}, \text{Na}, \text{H}) (\text{Al}, \text{Mg}, \text{Fe}, \text{Zn})_2 (\text{Si}, \text{Al})_4\text{O}_{10} (\text{OH})_2 - x \text{H}_2\text{O}$, where x -represents the H_2O content [43]. In general, the polymer matrices used to prepare PNC are hydrophobic whereas the nano clay is hydrophilic this opposing characteristic property makes it difficult to incorporate the nano clay into the polymer matrices. The formation of polymer clay nanocomposites depends upon certain factors like the nature of the clay used, the pH and polarity of the reaction medium, the nature of the solvent

Table 2: Various polymer/clay nanocomposites with their enhanced properties.

Polymer matrix	Nanoclay	Property enhanced	References
Polystyrene	Montmorillonite	Thermal stability	[46]
Polyamide	Montmorillonite	Tensile strength and elastic modulus	[47]
Polypropylene	Montmorillonite	Mechanical property	[48]
Styrene butadiene rubber	Montmorillonite	Flame retardancy	[49]
Poly(lactic acid)	Kaolinite	Gas barrier properties	[50]
High-density polyethylene	Kaolinite	Improvement in gas barrier properties	[51]
Polyethyleneimine (PEI) and anionic poly(acrylic acid)	Montmorillonite	Corrosion resistant	[52]
Polystyrene	Montmorillonite	Thermal stability and elastic modulus	[53]

used, the amount of the clay dispersion, the nature of adhesion between the polymer matrix and the nano clay and the interfacial attraction between the polymer matrix and the dispersed nano clay so achieving adequate dispersion and robust interfacial adhesion between nano clay and the polymer matrix is quite challenging and to overcome this and to achieve uniform dispersion and strong interfacial adhesion the nano clay are modified by ion exchange reaction methods in which the inorganic ions are replaced by organic surfactant molecules.

During this process, alkylammonium cations replace sodium or calcium cations, increasing the spacing between

the layers and providing additional space for the polymer chains to penetrate as well as increasing the hydrophobicity of the clay layer. As a result of this process, the hydrophilic silicate surface is converted to organophilic, weakening the polar contact between adjacent clay layers and increasing the hydrophobicity of the clay layer [45]. The literature has revealed that three types of structures (Figure 2) can be obtained during the synthesis of PNCs: phase separated, intercalated, and exfoliated.

3.4 Carbon-based nanoparticles

Carbon-based nanoparticles like carbon nanotubes (CNT), fullerenes, nanodiamonds, graphene, and graphene-based derivatives have created a lot of buzz and have gained a lot of importance among the scientific community because of their superior properties like large specific surface area, good mechanical properties, unique thermal and electrical conductivity which leads to their usage in a widespread range of applications and the combination of these carbon nanoparticles with the various polymeric matrices like thermoplastics, thermosetting, biopolymers, etc., will lead to new polymeric nanocomposites with enhanced functional and structural properties with applications in electronics, energy storage, biomedical, automobile, and aerospace [54, 55]. The high surface area, good biocompatibility, low toxicity, low cost, and more environmentally friendly synthesis methods of CNPs have led to their widespread adoption in a variety of scientific and technological applications. These qualities made them suitable for use in biological imaging, biomedical, photocatalysis, optical, and chemical sensing, in addition to their superb optical characteristics [56]. A brief overview of some of the important carbon-based nanoparticles is explained in the sections below.

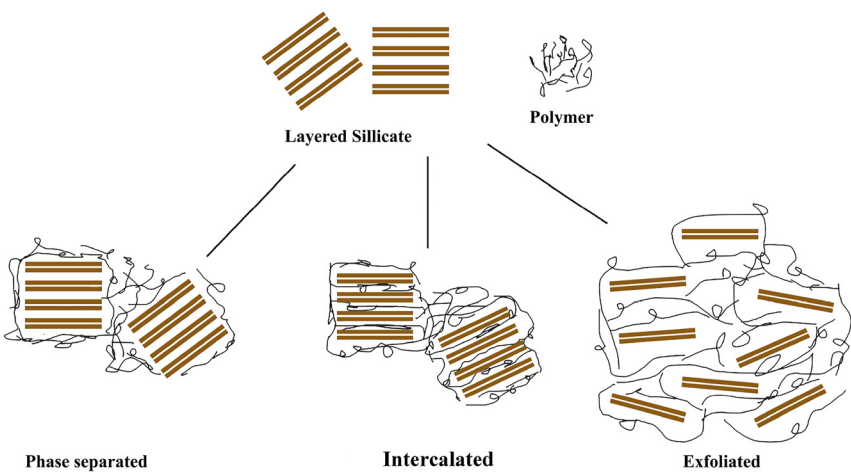


Figure 2: Different structures of polymer clay nanocomposites [44].

3.4.1 Carbon nanotubes

Carbon nanotubes have garnered a lot of attention in recent years because of their distinctive potential applications. Nanotubes can be either metallic or semiconducting, or semimetallic depending on their helicity and diameter, and they have a high elastic modulus and tensile strength. Since the carbon-carbon bond in graphite is one of the strongest in nature, carbon nanotubes make good candidates for strong, rigid structures. Carbon nanotubes have been shown to have a very high thermal conductivity that is significantly higher than that of bulk nanotubes and graphite. Carbon nanotubes display many exemplary properties due to which they find a variety of applications. The two varieties of CNT are single-walled carbon nanotubes (SWCNT) and multi-walled carbon nanotubes (MWCNT), and they can be made by folding a graphene sheet or a stack of graphene sheets along a specific axis. The diameter of SWCNT generally ranges from 0.8 to 2 nm and they are very light with density ranging from 1.3 to 1.5 g/cm³ and they show excellent mechanical properties like a high elastic modulus (0.32–1.47 GPa) tensile strength of 10–52 GPa and also display good thermal conductivity which is greater than 6000 W/m K and also electrical conductivity around 10⁶ S/m [57] MWCNT have a diameter ranging from 5 to 20 nm which can exceed 100 nm sometimes and a superior elastic modulus ranging from 0.27 to 0.95 TPa, the tensile strength of around 11–63 GPa, and can carry electrical currents up to 109 A/cm² [58, 59]. Carbon nanotubes exhibit superior mechanical, thermal, electrical properties due to which they have become key components in advanced sensors, optoelectronic devices, energy storage devices, and high-strength composite materials. The comparison of properties between SWCNT and MWCNT is shown in Table 3.

Carbon nanotubes, as shown in Table 3, offer incredible opportunities in electrical, mechanical, thermal, and field emission properties. Electrical properties include electrically conductive composites for electrostatic dissipation, shielding, and conductive sealants, electronic materials and devices for conductive inks and adhesives, electronic

Table 4: CNT-based polymer nanocomposites.

Nanoparticles used	Polymer	Property enhanced	References
MWCNT + Fe ₃ O ₄	PVDF (Poly (vinylidene fluoride))	Thermal conductivity and electromagnetic shielding	[62]
MWCNT + Fe ₃ O ₄	PE (Polyethylene) + PVDF	Thermal conductivity and electromagnetic shielding	[63]
SWCNT	PMMA	Fire retardancy	[64]
MWCNT	Polyurethane (PU)	Hardness and elastic modulus	[65]

packaging, device, and microcircuit components, and energy storage for supercapacitors and fuel cells. Mechanical characteristics like high-performance fibres, wear-resistant coatings, and low-friction coatings, as well as high-performance composites and reinforced ceramic composites. Thermally conductive paints and coatings, as well as thermally conductive polymer composites, are examples of thermal properties. The first CNT-based PNC was reported by Ajayan et al. [60] Some of the reported work on CNT-based PNC with different plastic types and CNT with the improvement in the properties is summarised in Table 4.

3.4.2 Graphene

Graphene is a two-dimensional (2D) nanomaterial that is very unique because of its structure which in general is a monolayer of sp² hybridised carbon atoms in a sheet that is one atom thick. It has very high mechanical strength, high electron mobility, and a large surface area due to which it has become the most commonly used reinforcement material in polymer nanocomposites [66]. The exceptional properties of Graphene are given below in Table 5.

Graphene is the ideal nanofiller to improve many properties like mechanical, thermal, and electrical in

Table 3: Property comparison of SWCNT and MWCNT [61].

Property	SWCNT	MWCNT
Mechanical strength	50–500 GPa	10–60 GPa
Specific gravity	0.8 g/cm ³	1.8 g/cm ³
Electrical conductivity	10 ² –10 ⁶ S/m	10 ³ –10 ⁵ S/m
Thermal conductivity	6000 W/m K	2000 W/m K
Thermal stability	>600 °C	>600 °C
Elastic modulus	~1.4 TPa	~0.3–1 TPa

Table 5: Unique properties of graphene [67, 68].

Property	Value (approx.)
Mechanical strength	130 GPa
Elastic modulus	1.0 TPa
Surface area	2360 m ² g ⁻¹
Thermal conductivity	5000 W m ⁻¹ K ⁻¹
Electron mobility	15000 cm ² /V s
Electrical conductivity	10 ⁸ S m ⁻¹

polymer nanocomposites. Through efficient interfacial stress-transfer, graphene might demonstrably improve the mechanical characteristics [69] of polymers. Graphene also can effectively toughen fragile polymers [70] by lengthening the nanocomposite's crack propagation route. By creating a conductive network for free electrons, graphene's superior electrical properties can undoubtedly increase the electrical conductivity of polymers [71, 72]. To create high-performance thermal conductive nanocomposites used in thermal management for high-power density devices, graphene's unique thermal conductance property is explored [73]. Graphene derivatives (such as reduced graphene oxide, rGO) can absorb visible light or near-infrared (NIR) wavelengths and transform light energy into phonon energy which leads to their usage in the manufacturing of photothermal nanocomposites such as actuators and solar steam generators [74–76]. The properties of PNCs are significantly influenced by the compatibility and interfacial interaction of the graphene with the polymer. Owing to its features which are generally anisotropic the distribution of graphene in polymer matrix is crucial in establishing the characteristics of nanocomposites. Graphene's strong van der Waals force causes it to aggregate and readily clump in the polymer matrix, degrading the properties of nanocomposites. To overcome these limitations, graphene is functionalized, and the functionalized graphene is used as a nanofiller [77–79]. The functionalization of graphene is carried out by various methods like covalent functionalization, non-covalent functionalization, and grafting. Since it increases solubility without changing extended π -conjugation, non-covalent functionalization has many advantages over covalent functionalization. In contrast, the covalent functionalization results in the formation of sp^3 defects on the graphene layer, which lowers the electronic conductivity of the material [80–82]. Various researchers used methods like spray deposition, vacuum infiltration, solution casting, melt mixing, and *in-situ* polymerization to synthesize graphene-based polymer nanocomposites. Every method of processing has benefits and drawbacks of its own. The features of the polymer matrix and nanofiller are taken into consideration while choosing the processing path and a brief overview of the reported literature on graphene-based PNCs has been highlighted in Table 6.

3.4.3 Carbon dots (CDs)

The development of CDs is more recent among the several types of carbon-based nanoparticles. The typical size of these zero-dimensional (0D) photoluminescent nanocarbon is less than 10 nm. However, other publications have also

Table 6: Graphene-based PNCs.

Polymers	Nanoparticles	References
Polystyrene	Graphene	[83]
Acrylonitrile-butadiene-styrene (ABS)	rGO	[84]
Poly (lactic acid) (PLA)	Graphene	[85]
Polycarbonate	Graphene	[86]
Polyethylene	Graphene	[87]
Polypropylene	Graphene	[88]
Syndiotactic polystyrene (s-PS)	Graphene	[89]
Poly (vinyl chloride) PVC	Graphene	[90]
Polytetrafluoroethylene (PTFE)	Graphene	[14]

shown that CDs can be much larger, i.e., up to 60 nm. The sp^2 and sp^3 carbon atoms that makeup CDs' structure have several polymer chains or functional groups linked to their surfaces [91–93]. The significant and exceptional properties of CDs, such as their exceptional electronic conductivity, photobleaching, and photoblinking properties, high photoluminescent quantum yield, fluorescence property, photo-decomposition resistance, changeable excitation, and emission attributes, increased electro-catalytic activity, good aqueous solubility, excellent biocompatibility, long-term chemical stability, and a very high surface area to volume ratio have attracted a great deal of attention from researchers [94–96] because of the numerous functional groups that are present on their surface, CDs are easily functionalized and also show outstanding sensing characteristics like particular, selective, and multiplex detectability. The surface of CDs has an abundance of functional groups like amine, carboxyl, hydroxyl, etc., which contributes to their good solubility in aqueous solutions and makes them simple to functionalize with other nanoparticles [96, 97]. Due to variations in their degree of carbonization, CDs' basic sp^2/sp^3 carbon skeleton frequently exhibits an amorphous carbon form or graphite lattice. Additionally, depending on their intended use or application, CDs' physicochemical properties can be easily adjusted [98–100]. The combination of CDs with different functional compounds and materials to create composites has recently sparked an increase in research interest. The drawbacks of pure CDs can be significantly reduced by composites, which can also broaden their uses. To create composites, CDs have been added to polymer matrices, polymeric gels, and molecularly imprinted polymers [101]. Because of CDs' small size and numerous surface functional groups, they can interact strongly with polymers and can thus disperse uniformly throughout them. CDs/polymer nanocomposites have increased flexibility, durability, strength, and hydrophobicity.

The polymer matrix determines the fundamental structure and properties of the CDs/polymer nanocomposites. CDs can either be covalently connected to or blended in a non-covalently manner with the polymer matrix. The strength of the secondary contacts between the polymer and CDs and/or the potential for covalently attaching CDs are determined by the chemical structure of the polymer or thermoset resin, which will also have an impact on the cost of preparation and CD dispersion. The three methods currently used to create CD/polymer nanocomposites are physical mixing, chemical grafting, and *in-situ* growth [102]. In general, natural or artificial/synthetic products can be used as carbon precursors in the manufacture of CDs. When opposed to manmade items, natural resources have several advantages, including low cost and significant environmental abundance. Furthermore, CD preparation from natural precursors can convert low-value natural/biological waste into useful and significant materials. Before creating their composites with polymers, CDs were prepared using a variety of natural materials, such as inner cassava peels, mango peels, Acerola fruit, and *Cedrus deodara*, chitosan, starch, and CDs have also been prepared using a variety of synthetic chemicals like glucose, citric acid, ethanolamine, etc. [103]. A list of several precursors for the synthesis of CDs-based PNC is given below in Table 7. Sensors that can be used in biological and chemical fields, bioimaging agents, optoelectronic devices, electrocatalysts, energy devices, light-emitting devices, drug carrier technology, and cancer diagnosis are just a few of the applications for CDs.

4 Methods of preparation

4.1 Nanoparticles

The nanoparticles can be prepared using top-down or bottom-up methods. Bottom-up methods involve carefully

assembling atoms or molecules that have previously been derived from their constituents via chemical reactions. The top-down approach, on the other hand, physically breaks down their respective bulk materials into nanoscale dimensions. Surface chemistry, self-assembly principles, and the physicochemical interactions of nearby atoms all have an impact on this process. Both approaches are important for the preparation of nanoparticles, and each has advantages and disadvantages. It is important to note that the bottom-up method can produce a consistent composition, less defective, and more uniform nanoparticles, whereas the top-down method typically produces somewhat defective and polluted nanoparticles with variable composition. The bottom-up method also promises a better probability of obtaining nanoparticles with fewer defects, a more homogeneous chemical make-up, and better short- and long-range ordering. This is because the bottom-up method is primarily motivated by lowering Gibbs free energy, which results in the production of nanoparticles closer to a thermodynamic equilibrium state. Instead, a top-down approach is likely to cause internal tension as well as surface imperfections and contamination, which will have a significant impact on the material's properties. The preparation methods applied to synthesise nanoparticles play an important role in determining the final properties of the material, and thus the following problems must be overcome for the creation and processing [112] of nanoparticles and nanostructures:

- (a) Overcome the massive surface energy caused by a high surface-to-volume ratio or a large surface area.
- (b) Ascertain that all nanoparticles have the desired size, homogeneous size distribution, morphology, crystallinity, chemical composition, and microstructure, which collectively produce the intended physical attributes.
- (c) Prevent nanoparticles and nanostructures from becoming coarser over time through aggregation or Ostwald ripening. A simplified representation of the preparation of nanoparticles is shown below in Figure 3.

Arc discharge, chemical vapor deposition (CVD), laser ablation, electrodeposition, inert gas condensation, aerosol, plasma, mechanical milling, and other techniques are used as physical processes to produce nanoparticles, whereas the chemical techniques include methods like chemical reduction, hydrothermal, solvothermal, microemulsion, sol-gel, microwave, coprecipitation, sonochemical, etc. [113]. A majority of these techniques are employed to produce various nanoparticles. Here is a brief review of a few of these techniques.

Table 7: Precursor for CDs/Polymer nanocomposites.

CDs precursor	Polymer	References
Citric acid/urea	Polyvinyl alcohol	[104]
Citric acid/urea	Polystyrene	[105]
Citric acid/NH ₃	Polyvinyl alcohol	[106]
Glucose	Polyamide	[107]
Chitosan	Polyacrylonitrile	[108]
Inner cassava peels	Polyvinyl alcohol	[109]
Ethanol amine	Polypropylene	[110]
Acerola fruit	Polyvinyl alcohol	[111]

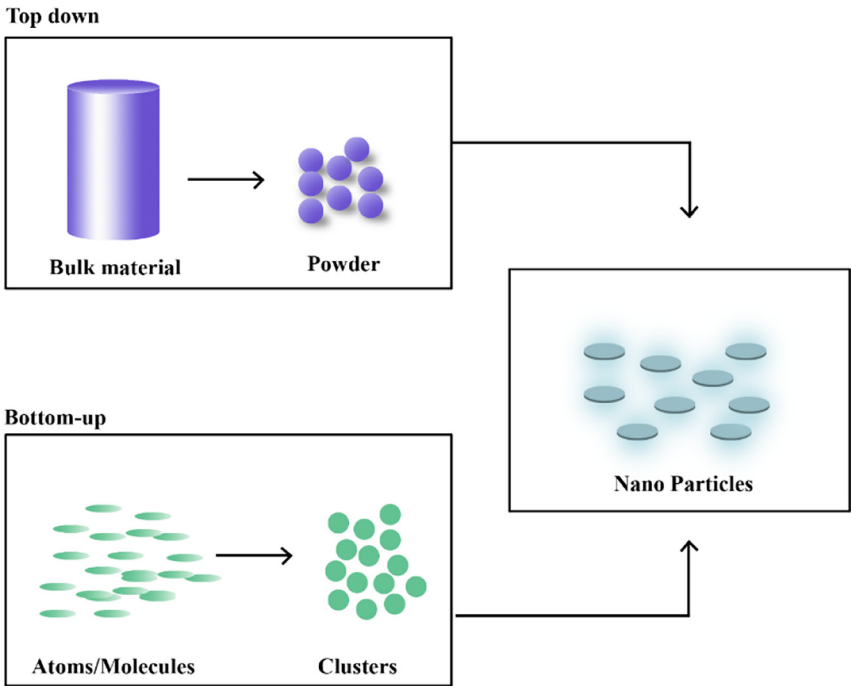


Figure 3: Approaches to nanoparticle preparation.

4.1.1 Physical processes

Different physical processes [114, 115] like arc discharge laser ablation, sputtering, molecular beam epitaxy, electron beam evaporation, thermal evaporation, pulsed laser deposition, ion plating, and others are examples of physical vapor deposition (PVD) techniques. The evaporation of the source material above its melting point in a vacuum chamber is the fundamental idea behind PVD because of the free path that the vacuum creates, the vaporised particles are deposited on a substrate.

In the laser ablation technique, the material is vaporised inside a vacuum chamber using high-power laser pulses, and the vaporised species are then condensed/cooled on a substrate to produce the required nanomaterial. The electrodes used, the presence of a catalyst, the voltage and current applied to the electrode, the pressure of the inert gas used, etc. are the determining elements for the final qualities of the nanomaterial that will be created during the arc discharge process. The arc discharge method is utilised to prepare CNT, metal and metal oxide nanoparticles etc. High mechanical shearing force is used in the high-energy ball milling process to break down the bulk material into nuclei, which are then produced during annealing to create the nanomaterial. Different milling systems, including high-energy ball mills, attrition mills, centrifugal mills, vibratory mills, low-energy tumbling mills, etc., were used in the mechanical attrition process.

Similar to this method, chemical vapor deposition involves heating the starting materials in the presence of an appropriate catalyst, which causes the precursor to change into nanomaterial at a relatively lower temperature. The electrodeposition technique can also be used to create other nanoparticles. A brief review of nanoparticles synthesized by different physical processes along with the size of the nanomaterial produced is given below in Table 8.

4.1.2 Chemical processes

Many bottom-up approaches are used in the chemical synthesis of nanoparticles. Among the techniques are wet

Table 8: Synthesis of nanoparticles by physical processes.

Nanomaterial	Method	Size obtained	References
ZnO	Horizontal oscillatory mill	20–30 nm	[116]
TiO ₂	Planetary ball mill	14–30 nm	[117]
CuO	Ultrasonic wave-assisted ball milling	20 nm	[118]
ZnO	Laser ablation	5–19 nm	[119]
Au (gold)	Laser ablation	9–15 nm	[120]
TiO ₂	Laser ablation	4–35 nm	[121]
SWCNT	CVD	1–5 nm	[122]
SWCNT	CVD	–	[123]
MWCNT	CVD	40–80 nm	[124]
MWCNT	CVD	–	[125]

chemical methods, precipitation, coprecipitation, sol–gel, solvothermal, hydrothermal, microemulsion, pyrolysis, and others. In the hydrothermal method, for example, a standardised dispersion of precursors in an aqueous medium is heated to high pressure and temperature in an autoclave to produce a variety of nanoparticles. This method is appealing for processing nanohybrids and nanocomposites, as well as for producing monodispersed, highly homogeneous nanoparticles [126].

The coprecipitation process is used to create a wide range of metal and metal oxide nanoparticles by adding an appropriate active agent in a predetermined concentration to an aqueous solution of the required metal ions under specific temperature and pH conditions. Under supersaturated conditions, processes involving nanomaterial nucleation, growth, and formation occur simultaneously. The properties of the produced nanoparticles were significantly influenced by Oswald ripening and aggregation. Co-precipitation is generally based on reactions that allow the simultaneous occurrence of nucleation, growth, and/or agglomeration processes. The supersaturated conditions are used to produce the insoluble products. Co-precipitation enables the creation of numerous small-sized particles in enormous quantities. Changes in particle size, shape, and aggregation may result from the secondary reactions. The main benefit of coprecipitation is its ease and speed of synthesis [127–129]. In the microwave method of synthesis of NPs, homogeneously dispersed precursors are subjected to microwave energies (900–2450 MHz), magnetic induction heating or ultrasound radiation (20–10 MHz) for a predetermined time to produce the desired nanoparticles. Ionic conduction and dipolar polarisation are typically used in the microwave process to heat materials. The control of the nucleation growth and, eventually, the properties of nanoparticles in this situation are greatly aided by the processing variables such as temperature, heating rate, precursor nature and concentration, etc. In the sonochemical technique, ultrasound waves are used which create transitory, highly confined zones with extraordinarily high temperature and pressure gradients. These abrupt alterations aid in the production of precursor nuclei, which are then converted into the appropriate nanoparticles [114].

Additionally, the sol–gel method is used to create metal oxide nanoparticles made of silica, titania, tin oxide, aluminum oxide, and hydroxide. A gel is a semisolid stiff mass formed by the creation of a chemical network from sol, and sol is a colloidal or molecular suspension of precursor solid particles in a liquid medium. For instance, the hydrolysis and polycondensation of metal alkoxides and halides to create a gel, followed by the removal of the liquid medium and calcination (thermal heating), results in the formation of the required metal oxides or hydroxide nanoparticles. The key benefits of the sol–gel approach include its

straightforward process, low cost, uniformity, and excellent purity of the resultant nanoparticles [130, 131]. The hydrothermal method is often used to synthesize metal oxide nanoparticles [132, 133] and in recent times it is used extensively to synthesize carbon dots [134–137]. Hydrothermal approach, which works by directly heating aqueous solutions that contain the desired natural/synthetic source. Dehydration, polymerization, passivation, and carbonization are the four key processes that can all be handled by this one procedure. This approach is extensively used since it produces non-toxic CDs and is easy, affordable, and energy-efficient. The hydrothermal method has a wide range of chemical compositions, very narrow particle size distributions, lower sintering temperatures, unitary step processes, densely sintered powders, sub-micron to nanoparticles with a narrow size distribution using simple equipment, lower energy requirements, quick reaction times, and the shortest residence time. A microemulsion process, which uses water (polar phase), hydrocarbon liquid, or oil (nonpolar phase), as well as a surfactant and cosurfactant mixture, can also be used to create nanoparticles. It is an isotropic, macroscopically homogeneous, and thermodynamically stable nano dispersion. Although the microemulsion technique is one of the most effective methods for producing inorganic nanoparticles, the process by which nanoparticles form in microemulsions is still unknown. Some scientists, however, have proposed a mechanism for the formation of nanoparticles in microemulsions. After mixing the reactant-containing microemulsion material, reactant exchange occurs during the collision of water droplets in the microemulsion. Because the reactant exchange occurs too quickly, the precipitation reaction occurs in the nanodroplets. This precipitation reaction is followed by the nucleation, growth, and coagulation of primary particles, resulting in final nanoparticles encircled by water and/or stabilised by surfactant.

The microemulsion's thermodynamic stability extends the formulation's stability [138] and shelf life. Microemulsions are made up of tiny droplets with a large interfacial area and are thermodynamically stable. These qualities make it easier to use them in the creation of minimally aggregated inorganic nanometer-sized particles. Formation of nanoparticles with a high specific surface area and frequently crystalline shape, a very high level of composition and particle size control are some of the advantages of the microemulsion method.

4.2 Processing of polymer nanocomposites

Processing methods have a substantial influence on the useful properties of polymer nanocomposites. The processing techniques can lead nanoparticles to aggregate and ultimately change their characteristics because of their high

surface area and huge characteristic aspect ratio. Additionally, by changing the distribution and degree of dispersion of the nanofiller phase, the processing method indirectly changes the bulk characteristics of polymer nanocomposites. Although numerous studies on the creation of nanocomposites have been published, it is still challenging to achieve homogenous dispersal of the NPs in a polymeric matrix. In the majority of cases, cluster and agglomeration formation is aided by the van der Waals attraction between nanoparticles [139, 140]. Additionally, there is an incompatibility between hydrophilic nanoparticles and hydrophobic polymers that leads to poor interfacial adhesion, poor dispersion, and poor characteristics. Consequently, nanocomposites occasionally demonstrate worse properties than traditional polymers, which restrict their useful application. Therefore, the scientific community has been paying close attention to synthesis procedures as well as the creation of novel integrated approaches to have greater control over both chemistry and morphology to solve these interfacial challenges. The primary goal of the synthesis of PNCs is to prepare them in a way that maximises the number of interfacial interactions between the polymer and nanoparticles. This can only happen if the nanoparticles are evenly distributed throughout the polymer matrix and the resulting polymer nanocomposite generates the highest degree of nano-dimensional interphases. Typically, mechanical shearing and ultrasonic forces are used for this purpose. But a key requirement for this purpose is the existence of suitable functional groups in both the polymer matrix and the nanoparticles. The type of nanofiller itself is at least somewhat related to the choice of processing method and its parameters (such as the kind of solvent employed,

temperature, mixing speed, time, etc.). For instance, graphite nanoparticles require particular solvents since they are very hydrophobic and difficult to dissolve in common organic solvents. Processing techniques can be divided roughly into kinetic and thermodynamic groups. While thermodynamic approaches rely on mediating interfacial contacts between the nanofillers to minimise the attractive forces and aid in dispersion, kinetic methods utilise an external energy source to disperse nanofillers into a host matrix [141]. To fully realise the potential of polymer nanocomposites, the process of preparation is extremely critical. In the laboratory and in industry, polymer nanocomposites are typically prepared using solution mixing, *in situ* polymerization, and melt mixing processes. According to their needs and applicability, additional techniques, such as sol-gel, templates, microemulsion, etc., are also used. Therefore, a brief description of a few methods like solution mixing, *in situ* polymerization, and melt mixing processes for the preparation of polymer nanocomposites is provided below.

4.2.1 Solution mixing

The solution mixing method, which is based on Stoke's law, is one of the easiest and most popular methods used for the synthesis of polymer nanocomposites (PNCs) [142]. This method involves the homogeneous dispersion of nanoparticles (NPs) in a polymer solution by using energetic mixing (agitation), which can be attained with the help of sonication, magnetic stirring, reflux, and shear mixing methods. The agitation step is then followed by the casting of the composite film by the evaporation of the volatile solvent [143]. A schematic illustration of the solution mixing method is displayed in Figure 4.

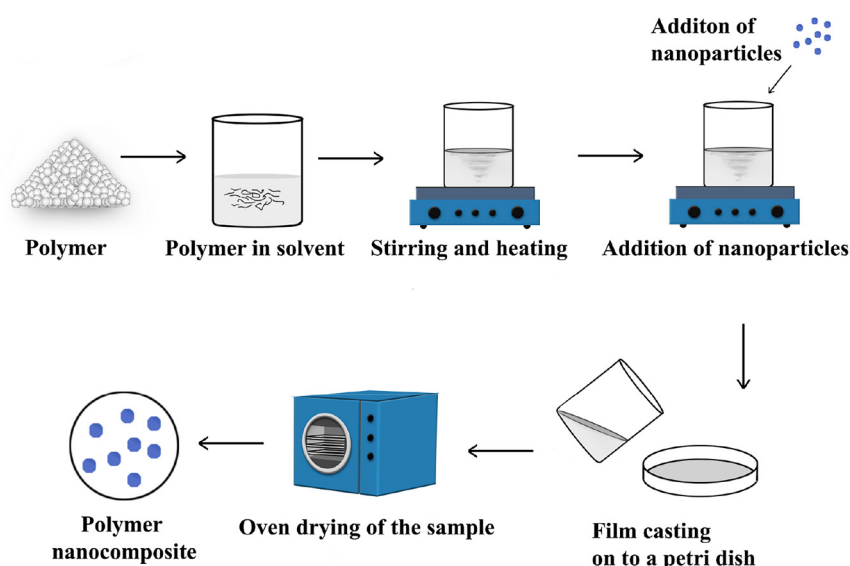


Figure 4: Schematic illustration of the solution mixing process.

Table 9: List of polymers with the solvents in which they are soluble [145].

Polymer	Solvent	Temperature (°C)
Poly vinyl chloride (PVC)	Tetrahydrofuran (THF)	25
Polyethylene glycol (PEG)	Water	25
Polyethylene	1,2,4-Trichlorobenzene (TCB) + 0.015 % butylated hydroxytoluene (BHT)	160
Polyisoprene	Tetrahydrofuran (THF)	40
Polyisoprene	Toluene	50
Polyethylene glycol (PEG)	DMF (dimethylformamide) + 0.1 % lithium bromide	50
Polyethylene terephthalate (PET)	2-Chlorophenol	100
Poly(lactic acid)	Tetrahydrofuran (THF)	40
Polypropylene	1,2,4-Trichlorobenzene (TCB) + 0.015 % butylated hydroxytoluene (BHT)	160
Polystyrene	Tetrahydrofuran (THF)	25
Polyurethane	Tetrahydrofuran (THF)	25
Polypropylene	o-Xylene	140

The solution mixing method is based on a solvent system that depends on the removal of the solvent from the solution by evaporation [144]. In this procedure, a polymer is dissolved in a suitable solvent, and a nanofiller is dispersed in the same or a different compatible solvent. The filler suspension and polymer solution are then commonly mixed using high-speed shear mixing, ultrasonication, or simple stirring employing a magnetic stirrer. Once the solvent has evaporated, the composite film is cast. The use of suitable solvents in this method depends upon the solubility of the polymer in that particular solvent. The commonly used solvents in this method include acetone, chloroform, toluene, xylene, cyclohexane, tetrahydrofuran (THF), dimethyl sulfoxide (DMSO), dimethyl formamide (DMF), etc. Some common polymers, along with the solvents in which they are soluble, are listed in Table 9 [145].

In this method, a polymer and well-scattered nanofillers are mixed together with the help of mechanical shearing and ultrasonication. The thermodynamics of mixing and interactions among the ingredients in the final nanocomposite determine the degree of intermixing between polymer chains and nanomaterials. To achieve uniform dispersion of nanoparticles in the polymer matrix, the interactions between nanomaterial and polymer chains must be stronger than the interactions between nanomaterial and solvent. Strong interactions between polymers and nanoparticles

enable polymer chains to penetrate the internal structure of the latter and soak onto their surface, producing the necessary nanocomposite [146]. The nanocomposite must be thermodynamically stable, which is controlled by advantageous entropic and enthalpic variables. Entropy goes down because the stiff solid nanoparticles stop the polymer chains from moving. However, entropy is almost nonexistent because the conformational motion of the nanoparticles goes up as the distance between layers goes up. The matrix and nanomaterial interact strongly, which causes the enthalpic loss to be much larger, but ultimately Gibbs' free energy turns out to be more advantageous [147]. Solution casting is a viable, simple, and cost-effective procedure because it doesn't need a complex experimental setup, excessive amounts of chemicals, or additional energy inputs. This process produces a large quantity of polymer nanocomposites in a short amount of time. The low viscosity of the solution used in this procedure has an advantage over other ways of creating polymer nanocomposites because it enhances the dispersion of nanoparticles in the matrix phase. The dispersion of the nanofiller in the polymer matrix is significantly influenced by processing parameters such as the weight per cent of the nanofiller employed, the amount and nature of the solvents used, and the time and type of sonication used. The choice of solvent is crucial to the effectiveness of the process since it must not only allow the complete dissolution of the polymer matrix material but also maintain the right viscosity for sonication-assisted uniform dispersion of nanoparticles. If there is an accumulation of leftover solvent at the interface between the nanofiller and the polymer, the physical characteristics of the composite, such as its glass transition temperature (T_g), are also impacted. Additionally, choosing the right solvent is crucial for improving the mechanical performance of the composite. However, there are several drawbacks to this method, including its high cost, flammability, potential health, and environmental risks brought on by the use of hazardous organic solvents, a dearth of interactions, etc. However, using a premade commercial polymer directly makes the solution process straightforward. It would also be advantageous to use water as the solvent [113, 114]. This method was used to create polymer composites using a range of nanofillers in host matrices like polyethylene (PP), polystyrene (PS), polycarbonate (PC), polyimide (PI), polymethyl methacrylate (PMMA), and polyacrylamide [148–150].

4.2.2 *In-situ* polymerization technique

Diffusing nanoparticles in a monomer—typically in the form of liquid resin—before the polymerization step is the first

step in *in-situ* polymerization. Given that the final nanocomposite contains both free chain and grafted nanofiller polymers, *in-situ* polymerization affords improved interfacial characteristics between the nanofiller and the host polymer matrix.

This method creates a polymer nanocomposite by generating the polymer in the presence of a well-dispersed nanomaterial. The process of preparing the well-dispersed nanomaterial begins with distension and diffusing it in the low-viscosity monomer or prepolymer of the aforementioned polymer to ensure that the polymer will form between the nanoparticles. Several methods can be used to initiate polymerization (heat, use of an appropriate initiator, etc). This method allows for the high loading of nanofillers without aggregation of a polymer-grafted nanoparticle [151]. Organic modifiers can aid in the polymerization process and aid in the dispersion of the nanoparticles. It can be an alternate method for creating nanocomposites with insoluble or thermally unstable polymers.

This method can occasionally be used without any solvents [152]. Through this method, monomer or prepolymer molecules get adsorbed on the dispersed nanoparticles superficially; during the polymerization process, polymer chains aid in the delamination, unbundling, or dispersion of the nanomaterial. In the case of the *in situ* procedure, the appropriate preformed polymer has a lower viscosity than the monomeric precursors or prepolymers compared to the solution technique. As a result, the interactions between polymer chains and nanoparticles are typically quite strong. A consistent dispersion of the nanoparticles in the polymer matrix is thus accomplished since the dispersion is highly homogeneous. As a result, it is a preferred method for creating various polymer nanocomposites, particularly resins, because it requires little to no energy and doesn't require any solvents. Due to the superior performance of the resulting polymer nanocomposites, *in situ* polymerization is preferred, especially in lab settings. A schematic illustration of this method is given in Figure 5. The technique can be applied to thermoplastics, elastomers, etc. Generally speaking, nanocomposites made using this method exhibit

better thermomechanical, flame-retardant, and electrical capabilities with lower nanofiller loading [59, 141, 153]. Additionally, it is a method that may lead to products with improved performance. The advantages include high transparency [154], tunable particle shape [155], and good nanofiller interfacial adhesion. This method enables the use of both thermoset and thermoplastic polymers, as well as higher concentrations of nanofillers [156] without agglomeration, better final product performance, expansion to a solvent-free state, and covalent bonds between the functional groups of the nanoparticles and the polymer chains. The simplicity of agglomeration is a significant drawback.

4.2.3 Melt mixing

This method uses the melt processing set-up to manufacture the necessary PNCs by uniformly mixing nanoparticles in premade polymer matrices. The industry prefers the melt-mixing method because it does not involve the use of a solvent and does not subject the product to the effects of polymerization or crosslinking reactions. Without any additional processes or equipment, the industrial polymer processing approach can be employed right away. The method involves heating amorphous or semi-crystalline polymers in a mixing device (such as an extruder, a Brabender plastic order, an injection molding machine, or a kneader), melting semi-crystalline polymers, or softening amorphous polymers, and then adding the nanoparticles to the matrix while applying a mechanical shear force to achieve homogeneous dispersion. Melt mixing does not use organic solvents, making it environmentally friendly.

Modern industrial techniques like extrusion and injection molding are compatible with melt mixing. The melt mixing technique has gained popularity because it has the potential for industrial applications. The nanofillers are directly dissolved into the molten polymer when using the melt mixing technique. The pressure that the polymer imparts to the particles during mixing while it is in the melt state depends on the molecular weight and weight distribution of the polymer.

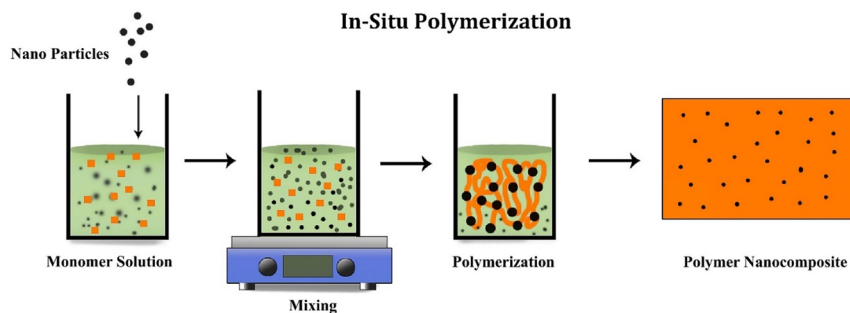


Figure 5: Schematic illustration of the *in-situ* polymerization process.

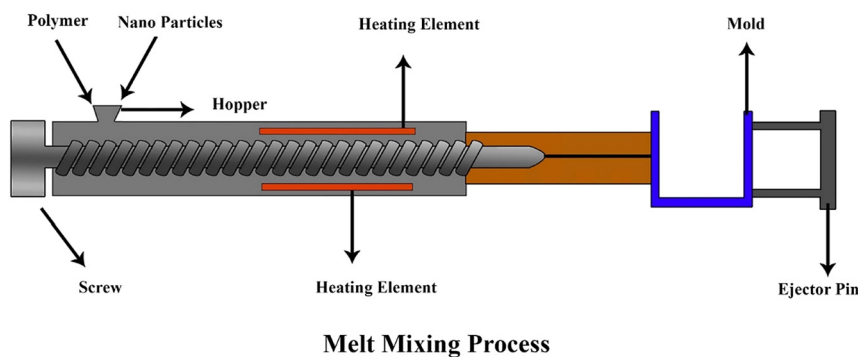


Figure 6: Schematic illustration of the melt mixing process.

The size of the agglomerates is reduced by high shear stress [156]. Large agglomerates initially disintegrate into smaller ones that are disseminated throughout the polymer matrix. This process, which is fundamentally dependent on time and the chemical affinity between the polymer and the surface of the nanoparticles, causes increased shearing, which separates the new agglomerates into individual particles [157, 158]. The two types of extruders—single and twin—are typically used for melt mixing though it should be emphasised that in some circumstances, high temperatures might have undesirable effects on the changed surface of the nanofiller, necessitating optimization. For this use, intermeshing co-rotating twin-screw extruders are extremely common. A schematic of melt mixing is shown in Figure 6.

The type of interaction between the polymer and the nanoparticles as well as the processing environment (temperature and residence time) are two parameters that are difficult to control in this method's disadvantages and most of the time, poorly intercalated nanocomposites are created because the melt viscosity of the premade polymer is often very high. As a result, sometimes it might be challenging to obtain well-dispersed nanoparticles. Polymer nanocomposites have been created by melt mixing with a variety of matrices, including polypropylene, PMMA, polyvinyl alcohol (PVA), PVC, polycarbonate, etc. Since extrusion and injection molding are two industrial processes that melt mixing works well with, they can be used commercially. Good nanoparticle dispersion, improved heat stability, enhanced mechanical qualities, low cost-effectiveness, and environmental friendliness (no usage of solvent) are a few advantages of this approach. The usage of high temperatures, which can harm the nanofillers' changed surface, is a significant constraint [159]. Overall, the process of creating polymer nanocomposites using this method is well recognised and has become popular because it is better from an environmental standpoint. In addition, this technique is particularly appealing for polymers that cannot be treated using solution procedures because of their insolubility in conventional solvents.

4.2.4 Other methods for the preparation of polymer nanocomposites

Apart from the methods for the synthesis of PNCs mentioned above, there are a few other methods for preparing polymer nanocomposites that are briefly discussed here. PNCs are also created using the sol–gel method, which is used to create nanoparticles. A polymer matrix is used in this process to create nanoparticles. As a result, the precursor or building blocks of the nanomaterial and the polymer matrix are present in an aqueous solution or gel that has been heated to a high temperature [160, 161]. During the process, the polymer aids in the growth and nucleation of the nanomaterial. The resulting nanomaterial is trapped in the polymer matrix. The majority of colloidal nanocomposites are created using this method, which has the disadvantage of requiring high processing temperatures, which may lead to polymer matrix degradation. Electrospinning is a technique for producing polymer nanocomposite-based nanofibers. This method of producing polymeric fibers is common. A high-voltage DC source, a microtip hydraulic automated syringe, and an electrically earthed collection are used to accomplish this. The desired amount of nanomaterial dispersion is mixed with the polymer solution in a suitable polar solvent or polar polymer melt using mechanical shearing followed by ultrasonication similar to the solution technique, while the jet of polymer solution or polymer melt was just trying to come out of the needle tip but was unable to do so due to surface tension. The applied voltage needs to be strong enough to stretch the jet and create nano-dimensional fibers by forcing electrostatic forces to overcome surface tension. The polymer strands should harden as they go along toward the collection plate. The hydraulic pressure of the syringe, the flow rate, the tip diameter, the strength of the electric field, the tip-to-collector distance, and other spinning variables all affect the uniformity, shape, and size of the nanofibers. In addition, the collector's geometry determines the product's size and shape. To produce the desired electrospun nanofibers, the homogeneous polymer–nanomaterial

mixture is electrospun under the recommended conditions, which include the applied voltage of 15–30 kV, tip-to-collector distance of 10–20 cm, feed rate of 1–15 mL/h, relative humidity of 25–35 %, and temperature of 25–60 °C [162].

5 Different types of polymer nanocomposites

5.1 Polymer–metal nanocomposites

The metals that are nanoscaled have distinctive characteristics that make them appealing for a wide range of cutting-edge functional applications. It is important to note that due to quantum-size effects, metals' characteristics vary significantly when their size is reduced [163]. According to the quantum-sized effects (such as electron confinement and surface effects), the novel features seen in nanoscaled metals are size-related characteristics. Due to plasmon absorption, these quantum-size effects give rise to distinctive features such as ferromagnetism, superparamagnetism, chromatism, and photo- and thermoluminescence [164]. Strong restrictions on their use have been placed on them due to the difficulties in managing these exceedingly small materials and their propensity to aggregate because of their high surface energy. In this line, stabilising these nanoscaled metals while also giving the matrix unique capabilities is possible by embedding them in strong matrixes that are functionally tailored with the desired properties of nanometals [165], and here, the polymers are one of the most effective and flexible category matrices. Superior functional applications are enabled by metal nanoparticles' low coordination number, specific optical, magnetic, and electronic characteristics, significant reactivity of edge and corner atoms, and improved catalytic activity in nanocomposites with polymers. These unique properties of metal-based polymer nanocomposites created in the nanoscale regime are size-dependent and modifiable by changing the counterpart's dimensions. For advanced practical applications, metals and polymers combined in nanocomposites can offer greater and more synergistic performance than the components utilised separately [166].

Diverse metal nanoparticles with different polymer matrices are used to create metal-based polymer nanocomposites. The selection of metal salts, reducing agents, solvents, and other chemicals while generating metal nanoparticles is critical for achieving the desired features and functions in metal–polymer nanocomposites. To support and govern the initial nucleation and subsequent growth of the first nuclei in the reduction reactions, a range

of reducing agents, solvents, and other chemicals are required for the solution technique of generating metal nanoparticles. Sodium citrate, hydroxyl amine hydrochloride, H_2O_2 , citric acid, carbon monoxide, formaldehyde, aqueous methanol, sodium carbonate, and sodium hydroxide are some of the reduction reagents. However, the use of UV light, microwave and sonochemical reduction, radiolysis, and plant extracts avoids the usage of the aforementioned reducing agents and addresses the growing concerns about the need for an environmentally friendly and long-lasting method of producing metal-based nanoparticles. Thiols, amines, surfactants, and other substances are examples of capping agents. Different polymers, such as poly(vinyl alcohol), sodium polyacrylate, polyethylene, and others, are included in the polymeric stabilisers. Various methods like *in-situ* polymerization, solution mixing, and melt mixing are used to prepare metal/polymer nanocomposites.

Metal nanoparticles have a large surface area to volume ratio, as well as an active surface, which results in several outstanding features. For instance, the main toxicity of nanosilver (Ag) for numerous bacteria is well established. Ag nanoparticles are more useful in biomedical and other environmental processes as a result of this feature. *In-situ* polymerization is one of the synthetic processes used to create polymer/Ag nanocomposites. A scholarly paper on the advancement of AgNanomaterial synthetic processes, factors influencing the growth, and PNC performance was published by Abbasi et al. [167] by *in-situ* polymerizing aniline, Wankhede et al. [168] created conducting polyaniline/silver nanoparticles (PANI-Ag) in this sector. Another study, Huang et al. [169] developed ferroelectric PNCs with inclusions of silver (Ag) nanoparticles to assess their dielectric characteristics and thermal conductivity. Gold nanoparticles coated with polymers have recently demonstrated a lot of promise for use in advanced material applications [170]. Optoelectronics, biosensing and imaging, drug administration, photothermal therapy for cancer treatment, and surface-enhanced Raman spectroscopy are some uses of gold nanoparticles [171]. Polymer/gold nanocomposites exhibit a variety of unique properties, including reduced light scattering, color change, and transparency, thanks to their adjustable stability, low intrinsic toxicity, biocompatibility, and plasmon resonance effects [172]. Different Au-CNT nanocomposites were represented [173] in terms of their synthesis and uses. The preparedness, properties, and possible applications of the smart core/shell nanocomposites comprising intelligent polymers and gold nanoparticles were also assessed. In a study [174] tetrachloroauric acid was used as an oxidant together with gold seeds to form *in situ* polymerized pyrrole and polypyrrole nanocomposites (AuNP/Ppy).

To create a hybrid nanocomposite resin with directly exploitable stereolithography (SL) procedures, Abdelrasoul et al. [175] combined AuNPs, polypropylene fumarate (PPF), and a biodegradable and photocurable polymer resin. To boost the particles' colloidal stability in this study, polyvinyl pyrrolidone (PVP) was applied to the AuNP nanofillers. *In situ*, polymerization of aniline in the presence of nickel oxide was employed by Jakeer Hussain et al. [176] to create poly-aniline/nickel oxide (PANI/NiO) composites (NiO). In a different work [177], NiFe and Fe nanoparticles were mixed in epoxy using various volume fractions to create nanocomposites. An impedance analyzer was used to examine the composites' permeability, permittivity, and loss tangents, among other characteristics. The outcomes were connected to the particle's saturation magnetization and field anisotropy, among other magnetic properties. More recently polymer-metal nanocomposites have found their way into diverse fields and they are used as effective agents in wastewater treatment because of their unique properties like heavy metal removal, photocatalytic degradation of organic dyes, and antifouling properties etc., they are also used in the cancer therapy. Una Bogdanovic et al. [178] have shown that Cu/PANI nanocomposite acts synergistically against three strains of bacteria, *Escherichia coli*, *Staphylococcus aureus*, and *Candida albicans* and the nanocomposite exhibited enhanced antibacterial activity and it can be used as a water disinfectant. Similarly, Kuo et al. [179] have synthesized Au nanorods/poly(styrene-alt-maleic acid) which are used in photodynamic therapy and also as hyperthermia agents to destroy cancer cells, and their study also proved that the prepared composites can be used as optical contrast agents to monitor cells by imaging in the near-IR region.

5.2 Polymer-metal oxide nanocomposites

Among the different classes of nanoparticles, metal oxide nanoparticles have been of quite significant interest both in the technological as well as scientific domains. When the metal oxide materials are reduced to the nano size range they display a variety of hybrid properties as compared to their bulk counterparts. With a definite electronic structure which can be either a semiconductor, metallic or an insulator depending on the type of structure, metal oxide nanoparticles can take on a variety of structural geometries. Metal oxide nanoparticles are the most diversified family of materials due to their unique properties, having applications in electroceramics, gas sensing, catalysis, superconductors, and energy conversions [180]. They have optical, electronic, electrical, magnetic, catalytic, and photoelectric properties that encompass virtually every aspect of

solid-state physics and materials science. These metal oxide nanoparticles can be spherical, triangular, star-shaped, or have nanowires, nanotubes, or nanorods, among other morphologies. Because of the high density and confined size, which led to intriguing discoveries in terms of physical and chemical characteristics, it is extremely desirable to appreciate the various aspects of metal oxide nanoparticle synthesis, features, and applications. The preparation of these polymer nanocomposites with different polymer matrices improves a number of characteristics, such as thermal stability, toughness, glass transition temperature, optical, tensile strength, and so on. The addition of these metal oxide nanoparticles enhances the characteristic properties of polymer nanocomposites. For the formation of polymer-metal oxide nanocomposites with a nanophase-separated structure, homogeneous dispersion of metal oxide nanoparticles and a reduction in the size of the polymer-metal oxide interface are essential because they fundamentally alter the physical properties of the nanocomposites. To create a homogeneous nanophase separation structure, a metal oxide must be uniformly distributed throughout the polymer matrix. The volume and weight fraction of the filler, the morphology of the system, and the structure of the interphase at the component interface all have an impact on the properties of these nanocomposites. An active area of research has been the blending of polymers and inorganic metal oxide nanoparticles; in particular, the engineering of flexible nanocomposites has attracted a lot of interest due to their notable differences in electrical, thermal, mechanical, and magnetic properties from the bulk polymers and inorganic metal oxide.

In general, these metal oxide nanoparticles are synthesized by conventional methods like precipitation, co-precipitation, hydrothermal, sol-gel, and ball milling those are used to synthesize nanoparticles, and several techniques, such as solution and melt mixing, sol-gel processing, growing polymers in the presence of nanoparticles, and *in-situ*, synthesis of metal oxides in the polymer matrix, can incorporate metal oxide nanoparticles into the polymer matrices but the interfacial interactions between the polymer matrix and the metal oxide nanomaterial determine the final property of the prepared nanocomposite. Since the metal-oxide nanoparticles have a high surface area they tend to agglomerate which leads to the problem of improper dispersion of nanoparticles inside the polymer matrix that will lead to degradation of the inherent properties of the polymer and the desired property of the nanocomposite is also not obtained. So, to avoid this problem, the surface of the metal oxide nanomaterial is functionalized or modified by the use of certain coupling agents like amines, vinyl, epoxies, etc. [181], and these surface modifications will

lead to a very specific interaction between the polymer matrix and the nanomaterial which ultimately leads to a proper dispersion of the nanomaterial inside the polymer matrix. A lot of research work on polymer metal-oxide nanocomposites has been carried out and a lot of work is in progress. Now, various researchers have reported a wide variety of research that has been carried out and an iota of that is presented here in this article. Lu et al. created polyaniline (PANI)/Fe₃O₄ nanocomposites using an *in-situ* polymerization process, and the resulting composites demonstrated new conductivity as well as superparamagnetic behaviour [182]. Wu et al. developed a melt mixing method to create a poly (ethylene terephthalate) (PET)-silica (SiO₂)/polystyrene (PS) nanocomposite [183] and the differential scanning calorimetry results showed that upon the addition of 2 wt% SiO₂, the crystallization temperature of the nanocomposite is measured to be 205.1 °C which is 11.6 °C higher than that of PET and the optical studies showed that the light transmittance rate of the nanocomposite reached 87.9 %, compared to 84.2 % for PET. Hashimoto et al. [184] have synthesized high-density polyethylene (HDPE)/TiO₂ (silanated TiO₂) nanocomposite and the mechanical properties of the nanocomposite were compared with HDPE/unsilanated TiO₂ and the results showed that the elastic modulus of the silanated composite was 10 GPa and the unsilanated composite showed a value of 7.5 GPa. In a similar experiment, Wu et al. [185] have surface modified TiO₂ nanoparticles with phenyl dichloro phosphate and have used them to prepare an epoxy resin/TiO₂ nanocomposite and the addition of these surface-modified nanoparticles to the epoxy matrix have improved the fire retardancy and the thermal stability of the polymer matrix. Mallakpour et al. [186] synthesized polyvinyl chloride (PVC)/ZnO nanocomposites and they displayed very high transparency and improved mechanical properties as compared with PVC and the nanocomposites also exhibited excellent antibacterial activity against two bacteria species: *E. coli* and *S. aureus*.

Sengwa et al. [187] have prepared PMMA (poly methyl methacrylate)/ZnO, TiO₂, SnO₂ nanocomposites by solution mixing method using tetrahydrofuran as the solvent and the nanofiller concentration was from 1 to 5 wt% and they reported that at the same filler concentration, the UV-absorbance was found to be low for the PMMA/SnO₂ nanocomposites and moderately high for the PMMA/TiO₂ nanocomposites. The film thickness-dependent analysis verified that, for PMMA/TiO₂ composites of the same composition, increasing the film thickness greatly improved the UV-Vis absorbance and decreased the energy bandgaps and at a temperature of 30 °C, it was discovered that the dielectric permittivity of the PNC containing ZnO and SnO₂ increased noticeably as the concentration of these

nanocrystallite fillers increased, but that of the PNC containing TiO₂ nanoparticles decreased in comparison to that of the host PMMA matrix. In most recent work polyaniline (PANI)/indium tin oxide, nanocomposites were investigated for optoelectronic applications by Al-Bataineh et al. [188] and their investigation revealed that as the indium-tin-oxide nanoparticle (ITONP) concentration (0, 2, 4, 8, 16 wt%) in the PANI/ITONPs nanocomposite rises, the electrical conductivity also rises and the refractive index of the PNC increases from 1.52 to 1.58 with the increase in the ITONP concentration. Thus, metal oxide polymer nanocomposites can be described as promising multifunctional materials for UV shielders, optical sensors, and bandgap tuners in the development of different sophisticated optoelectronic and organoelectronic devices. Because of the wide range of tailorable behavior of the optical and dielectric characteristics with changes in nanofillers and their concentrations, they may also be employed as low permittivity nanodielectric insulators and substrates to construct next-generation flexible-type electronic devices.

5.3 Polymer-carbon nanomaterial nanocomposites

The growth of the commercial application of engineered carbon-based nanoparticles is continuously rising in the fields of modern technology, medicine, the environment, and agriculture; the unique properties of carbon-based nanoparticles have attracted the attention of researchers and industries, which has sparked the expansion and innovation of techniques for significant industrial production [189].

Carbon nanoparticles like carbon nanotubes (CNTs), fullerenes, and graphene exhibit various exceptional features [190]. Carbon nanotube and graphene-based composite materials have been proposed as potential future advances. Carbon nanocomposites have piqued the interest of many industries, including biology, due to their amazing structural dimensions and exceptional mechanical, electrical, thermal, optical, and chemical capabilities. Significant advancements in carbon nanocomposite technology in recent years, as well as the discovery of new nanocomposite processing techniques, have been made to improve the functional impact of nanotube and graphene composites by providing appropriate synthesis techniques and enhancing the production of various composites based on carbon nanoparticles. Carbon nanocomposites are used in a variety of industries, including aerospace, batteries, the chemical industry, fuel cells, optics, power production, space, solar hydrogen, sensors, and thermoelectric devices. In comparison to typical fiber

composites, the mechanical performance of carbon nanomaterial-based composites is discovered to be significantly influenced by Van der Waals force interfacial compounds [191, 192]. Composites made of polymers and carbon nanostructures have significantly better mechanical properties because of the matrix's strong interactions with the nanoscale structure and very wide interfacial area [193]. The elastic modulus and electrical conductivity of graphene- and/or CNTs-based polymer nanocomposites were investigated experimentally, and the results show that the interfacial average shear stress of these composites is in the 500 MPa range, which is significantly higher than that of traditional fibre composites. The studies also demonstrated that, while the nanocomposites' ultimate strength and strain to failure are commonly reduced, their elastic modulus is generally increased [194, 195]. In recent years, carbon dots (CDs) have attracted increased interest as a fantastic nanomaterial in polymer nanocomposites because of their highly fluorescent property, quantum dot size, and the presence of a significant number of oxygen functional groups. As a result, they offer various advantages over other carbon-based materials and conventional semiconductor quantum dots, such as high optical absorptivity, water solubility, functionalizability, photobleaching resistance, chemical stability, toxicity, and biocompatibility [99, 196, 197]. In general polymer-carbon nanomaterial nanocomposites are prepared by the conventional methods like solution mixing, *in-situ* polymerization and melt mixing, etc., and some of the work that's has been reported has been highlighted here. Nanofiller concentrations of 0, 0.25, and 0.5 wt% were used by Jeon et al. [198] to build a CNT/epoxy coating. According to their findings, the coating's adhesive strength increased, and the impedance modulus for potential corrosion decreased. The fullerene- C_{60} -reinforced epoxy nanocomposite was examined by Liu et al. [199] using a range of C_{60} contents from 0.25 to 1.0 wt%, and the C_{60} /epoxy nanocomposites demonstrated better tribological and corrosion resistance. Dhanraj et al. [200] have prepared fullerene (C_{60})-PMMA composite and have conducted the bandgap study of the prepared composite and their studies revealed that with the addition of fullerene C_{60} , average absorption and band gap energy dropped while Urbach energy increased. Xavier et al. [201] have synthesized PVA/carbon nitride quantum dots (CN-QD) nanocomposites and have studied their optical properties and their results showed that the optical transparency diminishes as CN-QD concentration rises, the prepared nanocomposite films are luminescent and these optoelectronic qualities make them an attractive candidate for real-world phosphor applications. Eskalen et al. [202] have synthesized PVA/TiO₂/carbon dots (CDs) nanocomposite films by solution mixing method with different CDs concentration and the prepared

nanocomposite film's structural, morphological, mechanical, thermal, and optical characteristics have all been thoroughly studied. When compared to PVA/TiO₂ polymer composite, PVA/TiO₂/carbon dots(CDs) nanocomposite melting point was higher, but as CD concentration rose, PVA's melting point steadily dropped. With the addition of TiO₂ nanoparticles, PVA's tensile strength reduced. They also investigated how well Rhodamine B degraded via photocatalysis using the PVA/TiO₂/CDs nanocomposite with increasing concentrations of CDs. Although the degradation rate was first accelerated by doping CDs, it was eventually seen that the degradation rate decreased as the amount of CDs increased. Zhang et al. [203] have tested the mechanical properties of polyvinyl acetate (PVAc)/CDs nanocomposite where the average particle diameters of CDs are 2.5 nm and 8.5 nm and they have reported that with a reduction in particle size, the relaxation time, elongation at break, of PNCs will increase while the tensile stress decreases.

In summary, the addition of various carbon-based nanoparticles to polymer matrices proves to be a viable method to enhance the polymer properties and it also results in materials with a set of improved properties that can prove effective in various applications. CNPs can be used to create materials that can be applied in a wide range of high-performance engineering applications. A number of variables, including the type of CNPs used, their properties, dispersion, alignment, purity, aspect ratio, functionalization, and how they interact with the polymer matrix at the interface, affect the quality of CNPs-reinforced PNCs. Therefore, it is crucial to concentrate on various functionalization methods for CNPs and to discover a way to increase their compatibility and dispersion with polymer matrix.

6 Characterization of polymer nanocomposites

PNCs are characterised using a variety of techniques, such as spectroscopic and microscopic methods to determine their structure, morphology, elemental details and methods like thermal and mechanical analysis are done to get detailed information on the T_g , hardness, impact strength of the PNCs [204]. The structure of the PNCs can be known by using transmission electron microscopy (TEM) and another technique namely wide-angle X-ray diffraction technique is used to describe the internal structure of polymer nanocomposites (WAXD). WAXD can also be used to examine the nanoparticles' dispersion within the polymer matrix. Transmission electron microscopy is used to characterise polymer nanocomposites because it allows for a qualitative

understanding of the internal structure, spatial distribution of the distinct phases, and direct inspection of the defect structure. The degree of dispersion of the nanoparticles in polymer nanocomposites is also investigated using scanning transmission electron microscopy (STEM) the surface morphology of the polymer nanocomposites is studied by using scanning electron microscopy (SEM) and many distinct forces, such as adhesion force, magnetic force, and mechanical characteristics of the PNC, are measured using atomic force microscopy (AFM). To identify functional groups and to comprehend the structure of the polymer nanocomposite Fourier transform infrared spectroscopy (FTIR) is used and in order to achieve an uniform dispersion in the polymer matrix nanoparticles are functionalized and the effects of this functionalization on the nature of the interaction between the polymer matrix and the nanoparticle is also studied by FTIR. Thermal analysis method like differential scanning calorimetry (DSC) is used to identify the type of crystallisation occurring in the matrix. Some of the nanoparticles added to the polymer matrix are luminescent and as a result of their addition the polymer nanocomposite shows luminescent properties [205] which are studied by fluorescence spectroscopy. When the nanofillers are added to the polymer matrix and if the dispersion is not uniform they lead to the formation of an adsorption layer which restricts the chain moment and this phase structure and the dynamics of the polymer chains can be studied by solid state nuclear magnetic resonance (NMR) spectroscopic techniques. Raman spectroscopy is one technique which is specifically used to characterize carbon nanoparticles-based polymer nanocomposites, many carbon nanomaterials like fullerenes, carbon nanotubes, graphene etc., can be characterized by Raman spectroscopy because even if they are disseminated in a polymer matrix at a very low content, they exhibit resonance-enhanced Raman scattering effects, giving rise to strong, well-defined bands [206].

7 Properties of polymer nanocomposites

The properties of the polymer nanocomposites depend on many important factors like the combination of the dispersed nanomaterial and the polymer matrix, the rate of dispersion of the nanofiller, the interfacial interaction between the matrix and the dispersed phase, the characteristic of the nanofiller and the polymer matrix and finally on the structure of the composite formed. Nanoparticles are highly reactive due to their large surface area and surface energy, making it very simple for them to respond

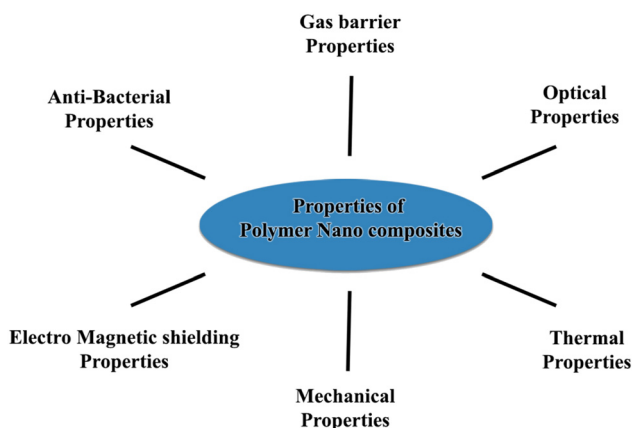


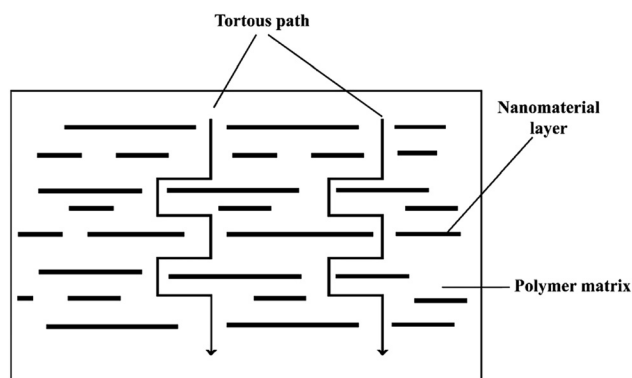
Figure 7: Properties of polymer nanocomposites.

differently to other chemicals and materials, despite the fact that these reactions involve different kinds of chemical and physical interactions. Consequently, one of the areas of current research in the field is the surface functionalization of nanoparticles. The enhancement of the characteristics of the individual polymers in their respective polymer nanocomposites is also greatly influenced by surface functionalization. Functionalization is used to incorporate multi- or polyfunctional groups on the surface of a conventional nanomaterial. It is important to note here that several unique qualities, including antibacterial, biocompatible, controlled medication administration, flame retardancy, etc., can also be obtained by appropriately functionalizing the right nanoparticles, so it may also be possible to use the same or comparable kinds of nanoparticles for various purposes by using appropriate functionalization or surface chemistry adjustment. Various properties of polymer nanocomposites are shown in Figure 7. And few properties like mechanical, thermal and gas barrier properties are explained briefly in this article.

Properties like mechanical properties get enhanced upon the addition of a very small amount of the nanofiller this is due to the fact that the excellent stiffness that the nanoparticles (NPs) possess aids in improving the tensile strength of polymeric matrices, which are often considered to be materials with low tensile strength. A strong interface is produced as a result of NP's capacity to achieve a superior bonding with the polymers which enables the smooth and simple transfer of stresses brought on by loading from the weak matrix to the strong reinforcements that leads to the formation of a polymer nanocomposite with greater mechanical strength [207]. The particle-to-particle contact inside the chosen polymeric matrix is one of many mechanisms that affect the composite's final mechanical property. This process has a significant impact on the mechanical

property. Due to the increased surface area of the NPs and the subsequent development of strong interactive forces of attraction between the particles, the tendency of the NPs to form agglomerated clusters/aggregates tends to negatively affect the mechanical properties of the PNCs by limiting the effective interfacial area [208, 209] and the agglomeration is also dependent on the shape and degree of orientation of the nanoparticles in the polymer matrix [210]. According to published data, the density and size of the nanoscaled fillers affect the interfacial area. The ultimate mechanical properties of PNCs are improved as a result of the nano-fillers' reduced size and density, which also increases the interfacial area [211]. The interphase thickness is seen to be equally as important in defining the tensile properties of the composite material as the particle size [212, 213]. This is so that the PNCs can sustain loads and delay the probability of failure since the connection between the NPs and the matrix is stiff in the interfacial region [214]. Some of the research that has been published emphasises how this important interface thickness is related to the amount of high molar mass polymer chains that are adsorbed onto the surfaces of NPs. This condition can be controlled by changing the loading of NPs [215]. It has been claimed that small-sized NPs do not improve the tensile strength of the composite material and instead create a thin and weak interphase. This is also valid for larger NPs that result in thicker interphase. Therefore, it cannot be argued that for the material to have the best tensile strength, particle size and interphase thickness should be coordinated. Smaller NPs with thicker interphases must be used in order to get a larger reinforcing effect and, consequently, an improved mechanical property. The final material property is also influenced by the volume/weight fraction of NPs in the chosen polymer matrix [216, 217]. According to a review of the literature, the type of matrix material utilised has a significant impact on the mechanical characteristics of PNCs that include high volumes of nanoscale particles [218]. In general volume/weight fraction of NPs is beneficial for elastomer/thermoplastic matrix whereas low volume/weight fraction of NPs is beneficial for thermosetting and this is mainly because in comparison to an elastomer, a thermoset resin has a much higher initial tensile strength and for elastomer matrices with a lower modulus, a high volume fraction of NPs provides the necessary densification by particle entanglement in polymeric chains. On the other hand, higher weight percent of nano-filler additions causes the production of particle aggregates of micrometer size in the majority of thermoset matrices, acting as stress risers in the nanocomposite and lowering the mechanical characteristics of the composite material. So, in order to accomplish the desired property of the PNCs, one must be careful not only in

the selection of the appropriate nanofiller and polymer matrix but also in determining the volume/weight percent that must be integrated. When the nanofiller is added in an excessively big or excessively small amount, the desired outcome might not be achieved. Achieving the desired material property appropriate for the particular application can be accomplished by adding the nanofiller in the right amount [219, 220]. In addition to mechanical properties, other properties like thermal and gas barrier properties also get enhanced with the addition of nano-sized filler to polymer matrices. In general, one of the main shortcomings of polymer matrices is their thermal stability which is very poor at elevated temperatures which can be significantly improved by the addition of nanoparticles and this is due to the fact that nanoparticles often operate as heat insulators, mass transport barriers for volatiles produced during the decomposition process, and modify the pathway of their decomposition, their integration increases the thermal stability. As a result of stiff nanostructures restricting the mobility of polymeric chains, the glass transition temperature (T_g) is likewise raised. The creation of a polymer nanocomposite may also increase the total enthalpy of crystallisation as well as the crystalline melting temperature (T_m). Nanoparticles serve as nucleating agents in these situations to promote the crystallisation of the crystalline polymer matrices [221–223]. Barrier properties of the polymers also get enhanced with the addition of nanoparticles and an enhanced gas barrier property is seen in the polymer nanocomposite and the reason behind this is the nanoparticles with a higher surface area obstruct the diffusion path of the penetrating gas molecules by creating a tortuous or zigzag path (Figure 8) leading to the improvement in the barrier properties and this enhancement is due to the following reasons (i) The incorporation of



Barrier enhancement property of Polymer Nano composites

Figure 8: Zig-zag path model for barrier enhancement properties of polymer nanocomposites.

the nanoparticles that are impermeable in the permeable polymer matrix lowers the available area for diffusion and (ii) The gas diffusion paths are lengthened by the tortuous path that gas molecules take as they travel around the impermeable nanoparticle filler [224]. So, in general, there are three key characteristics that define the polymer nanocomposites gas barrier performance: (1) Filler characteristics like shape factor, content, aspect ratio, dispersity, volume fraction, filler adsorption, and activity/resistance to gas diffusion, (2) core polymer characteristics like permeability, crystallinity, and intrinsic barrier property, and (3) dispersion quality in the composite film like filler distribution, orientation, exfoliation, intercalation, agglomeration, etc.

Table 10 shows some of the published work on the effect of NPs on the mechanical and thermal properties of the polymer nanocomposites.

8 Applications of polymer nanocomposites

As a result of their integrated structure, polymeric nanocomposites (PNCs) exhibit properties that none of their

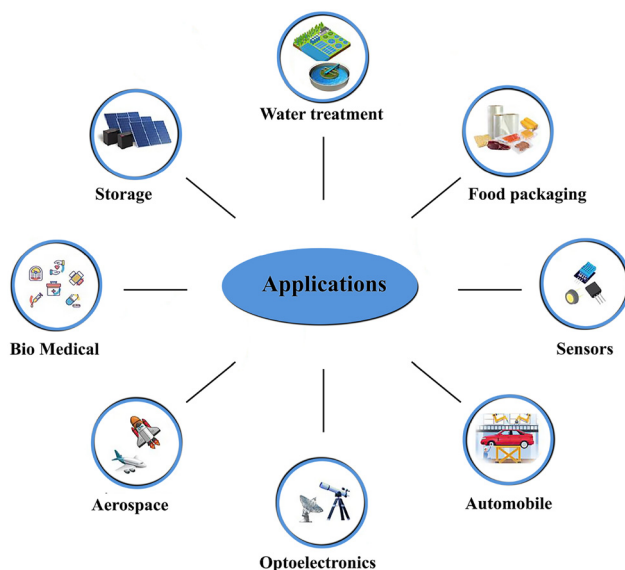


Figure 9: Applications of polymer nanocomposites.

individual constituent materials separately have, making them remarkably promising for a wide range of applications (Figure 9). The surface areas, magnetic properties, optical properties, and catalytic activity of PNCs can also be controlled, and they can be produced in any shape or size

Table 10: Property enhancement in polymer nanocomposites.

Nanoparticle	Polymer matrix	Processing technique	Weight percentage (wt.%) of nanoparticle	Property enhanced	References
Al ₂ O ₃	80–90 % Diglycidyl ether bisphenol A and 10–20 % aliphatic diglycidyl ether mixture	Solution mixing	1 %	Tensile strength and elastic modulus	[225]
Al ₂ O ₃	PMMA	<i>In-situ</i> polymerization	2 %	Tensile strength elastic modulus, impact strength	[226]
Graphene oxide	Polyimide	<i>In-situ</i> polymerization	3 %	Tensile strength	[227]
AgO	Bisphenol A	Sonication + addition of curing agent	1 %	Tensile strength and flexural strength	[228]
ZnO	Epoxy	Sonication + addition of curing agent	5 %	Increase in glass transition temperature and melting point	[229]
Boron nitride	PVA (polyvinyl alcohol)	Electrospinning	33.1 %	Thermal conductivity	[230]
Al ₂ O ₃	poly(vinyl pyrrolidone) (PVP)/poly(ethylene oxide) (PEO)	Solution mixing	1–5 %	Increase in melting point temperature and degree of crystallinity	[231]
Silica (SiO ₂)	Poly(vinyl pyrrolidone) (PVP)/poly(ethylene oxide) (PEO)	Solution mixing	1–5 %	Increase in melting point temperature and degree of crystallinity	[231]
Silica (SiO ₂)	Poly-methyl-meth-acrylate (PMMA) and polyvinyl alcohol (PVA)	Solution mixing	2 %, 4 %, 8 %, and 16 %	Thermally stable upto 110 °C	[232]
Boron nitride	Polyethylene	Melt mixing	1 %	Thermal decomposition temperature was increased by 12 °C	[233]

Table 11: Applications of polymer nanocomposites.

Matrix	Reinforcement	Weight percentage (wt%) of reinforcement	Major outcomes	References
1 Water treatment applications				
Polyvinylidene fluoride (PVDF) and polydimethylsiloxane (PDMS)	Multiwalled carbon nanotube	0.05 %	Improved surface roughness, hydrophobicity and antifouling properties	[234]
Poly (vinylidene fluoride) (PVDF)/sulfonated polyethersulfone (SPES)	TiO ₂	0.1–6 %	Enhanced the antifouling properties by changing the membrane surface from hydrophobic to hydrophilic	[235]
Polycarbonate (PC)/polyurethane (PU)	(Al ₂ O ₃)	0.5–2%	The PC-PU membrane's porosity and hydrophilicity as well as its antifouling and humic acid removal ability were improved by the addition of 1.5 wt% Al ₂ O ₃	[236]
Polyethersulfone (PES)	Nitrogen-doped porous graphene oxide (N-PGO)	0.05–1 %	The PNC prepared efficiently removes basic red –46 dyes	[237]
PVA/PAA/Fe ₃ O ₄ /MXene	Ag nanoparticle	0.25 %	Improved 4-nitrophenol and 2-nitroaniline elimination and reduction by catalysis	[238]
Aromatic polyamide (PA)	Multi-walled carbon nanotube (MWCN)	0.25–1.5 %	Significant elimination of NaCl and humic acids was achieved by increasing the MWCNT concentration in the composite membranes	[239]
<i>m</i> -Phenylenediamine (MPD) and trimesoyl chloride (TMC)	Graphene oxide quantum dot (GOQD)	0.1–1 %	Antifouling and chlorine resistance	[240]
Polyethersulfone (PES)	Graphene oxide		Enhanced the hydrophilicity and anti-fouling property of the membrane	[241]
2 Antimicrobial properties				
PLA & D limonene	Zinc oxide	3 %, 5 % and 8 %	5 wt% concentration of zinc oxide increased the nanocomposite's thermal stability, ductility, and antibacterial effectiveness against <i>Escherichia coli</i>	[242]
Polyaniline (PANI)/Polyvinyl alcohol (PVA)	Ni/Cu		Excellent antibacterial and antifungal action was demonstrated by the PNC	[243]
Polyvinyl alcohol (PVA)	GO/rGO	2 % and 4 %	Inhibitory action toward <i>Pseudomonas aeruginosa</i> , <i>Escherichia coli</i> , and <i>Campylobacter jejuni</i> has been shown in polymer nanocomposite with GO as dispersant (4 wt%). Using the well diffusion method, polymer nanocomposite containing rGO (4 wt%) has inhibitory effect against <i>Pseudomonas aeruginosa</i> , <i>Candida albicans</i> , and <i>Escherichia coli</i>	[244]
Polyurethane (PU)	Graphene oxide (GO) and reduced graphene oxide (RGO)	0.1–0.6 %	GO and RGO PU nanocomposite shown superior and effective antibacterial behaviour against <i>Staphylococcus aureus</i> and <i>Escherichia coli</i>	[245]
Poly methyl methacrylate (PMMA)	Graphene-silver nanoparticles (G-Ag NPs)	1 % and 2 %	The PNC showed an efficient antibacterial activity	[246]
Polyvinylidene fluoride (PVDF)	Graphene oxide-silver nanoparticles (GO-Ag NPs)	1 % GO + 0.5 %–2 % Ag Nps	The addition of AgNPs and GO to PVDF membranes results in enhancements of membrane wetting, mechanical strength, elastic modulus, water permeability, antifouling and antimicrobial properties	[247]

Table 11: (continued)

Matrix	Reinforcement	Weight percentage (wt%) of reinforcement	Major outcomes	References
Polyethylene glycol (PEG)	Ag–ZnO–reduce graphene oxide (rGO)		The PNC showed strong antibacterial activity	[248]
3 Optoelectronics applications				
Trimethylsilyl cellulose (TMSC)	CuInS ₂	2 %, 6 %, and 12 %	The composite displayed a favourable power conversion efficiency, which will lead to its use in solar cells	[249]
Polydisilanobithiophene (P-DSBT)	Zinc oxide (ZnO)	10–100 %	P-DSBT/ZnO hybrid nanocomposite shows a good dispersion of ZnO NPs throughout the polymer at 10, 20 and 40 wt% of ZnO which leads to their use in optoelectronic and gas sensing devices	[250]
Polyvinyl alcohol (PVA)	Nickel (Ni) NP	1.0 %, 3.9 %, 5.4 %, 6.7 %, 10.1 %, 13.4 %, and 17.2%	Optical band gap of the PNCs is reduced and optical dielectric constants and conductivity is increased with the increase in concentration of Ni NPs in the polymer matrix	[251]
Poly (vinyl pyrrolidone)/poly(vinyl alcohol) (PVP/PVA)	Alumina (Al ₂ O ₃) nanoparticles	1–5 %	When the Al ₂ O ₃ NPs' concentration in the polymer matrix grew from 0 wt% to 5 wt%, the energy band gaps fell from 5.17 to 5.10 eV while the Urbach energy as well as the refractive index of the PNC materials increased, validating their usefulness as better optical materials	[252]
Polyvinylidene fluoride (PVDF)	Carbon quantum dots (CQDs)	0 %, 1 %, 3 %, 5 % and 10 %	There is a drop in the energy band gap and significant improvement of the refractive index and DC electrical conductivity (DC) of the nanocomposite films when the wt% ratio of CQDs/PVDF is raised up to 10 wt%	[253]
Polyvinyl alcohol (PVA), polyaniline (PANI)	Ag NPs	3 % and 6 %	The improvement in the optical conductivity of composite materials by doping the PVA polymer with PANI and Ag NPs signifying that the prepared PNC is a promising material for optoelectronic applications	[254]
Polyamic acid (PAA) of poly (pyromellitic dianhydride-co-4, 4'-oxydianiline), amic acid (PM)	Silica (SiO ₂) NPs	0 %, 6 %, 8 %, 10 %, and 12 %	The increase in nanosilica concentration in the polymer matrix leads to a drop in the band gap (E _g), and, transmittance and an improvement in the refractive index, dielectric constant, optical conductivity and free charge carrier concentrations of produced composite films	[255]
Polyvinyl alcohol (PVA)–polyethylene oxide (PEO)	Magnesium oxide (MgO) NPs	2 %, 4 % and 6 %	The conductivity of the resulting nanocomposites increases with frequency, while the dielectric constant and dielectric loss decrease. The results show that the (PVA-PEO-MgO) nanocomposites are versatile enough to be used in a wide range of electronic applications	[256]
4 Sensor applications				

Table 11: (continued)

Matrix	Reinforcement	Weight percentage (wt%) of reinforcement	Major outcomes	References
PANI	Multiwall carbon nanotubes (MWCNTs)		High sensitivity to detect ammonia (92 % for 100 ppm), quick response time (9 s)	[257]
Polypyrrole	Nitrogen-doped multi-wall carbon nanotube		The polymer nanocomposite was annealed at 350 °C and it demonstrated an excellent linear response toward NO ₂ concentrations of 0.25–9 ppm at ambient temperature	[258]
Polypyrrole	Multi-walled carbon nanotubes (MWCNTs)/indium tin oxide (ITO)		The PNC drastically improved selectivity for Pb(II)	[259]
PANI	Multiwalled carbon nanotube-3-aminopropyltriethoxysilane		The PNC exhibited high selectivity towards Cd ²⁺ ions	[260]
Poly (aniline-co-indole)	Cu–Al ₂ O ₃	5 %	The PNC has excellent NH ₃ gas sensing	[261]
Polythiophene	Chitosan	10 %	The resulting PNC has good sensitivity and rapid response towards ammonia gas at ambient temperature	[262]
5 Electromagnetic shielding for aerospace applications				
Polymethyl methacrylate (PMMA)	Multiwalled carbon nanotubes (MWCNT)	10 %	The PNC showed an electromagnetic interference shielding effectiveness up to 40 dB (decibel) in the frequency range 8.2–12.4 GHz (X-band)	[263]
Nitrile butadiene rubber (NBR)	Fe ₃ O ₄	10–40 %	The PNC with 40 wt% Fe ₃ O ₄ had a higher EMI shielding effectiveness of about 92 dB in the frequency range 11–12 GHz	[264]
Polyurethane	MWCNT	10 %	The PNC showed an electromagnetic interference shielding effectiveness (SE) up to 29 dB (decibel) in the frequency range 8.2–12.4 GHz (X-band)	[65]
Polypropylene	Carbon black	1%, 3%, 5%, 7.5% and 10 %	The PNC with 10 wt% of carbon black had an optimum EMI SE of about 43 dB in the X-band	[265]
Ethylene methyl acrylate (EMA)	MWCNT	10 %	The PNC showed EMI shielding effectiveness of 25.1 dB with superior mechanical and electrical characteristics	[266]
Epoxy resin	Fe ₃ O ₄	15 %	The PNC showed total shielding efficiency was 36.6 dB at 8.2 GHz	[267]
Epoxy resin	Carbon black nanoparticle	50 %	Total shielding efficiency was 36.6 dB at 8.2 GHz	[267]
Low-density polyethylene	CoFe ₂ O ₄ /MWCNT/graphene		Total shielding effectiveness of about 49 dB was achieved at 10.3 GHz	[268]

that is required. PNCs have potential in a variety of environmental and industrial applications due to their low cost and light weight. Some of their diverse uses include packaging materials, energy storage devices, sensors, bio-medical applications, antimicrobial applications, wastewater treatment, catalysts, cosmetics, transportation and safety, defence systems, thermal flow control, information industry, novel catalysts, cosmetics, sports, etc. Just a few of the applications that

have been documented in the literature have been highlighted in Table 11.

9 Conclusions

This review paper comprises of the fundamentals, preparation, characteristics and applications of polymer nanocomposites

through a thorough analysis of the available, published research work. It can be seen that the best fabrication procedures for using NPs as fillers in PNCs are melt mixing, *in-situ* polymerization, and solution mixing approaches. The most important, yet most challenging, step in producing high-quality PNCs is the uniform dispersion and good compatibility of the NPs with the polymer matrix. The matrix material is made stiffer by the uniform distribution of the NPs, which also contributes to the improvement of the PNCs' mechanical and thermal properties. In the future, attempts can be made to promote hybridization by including multiple types of NPs in the polymer matrix which can create hybrid PNCs and research needs to be carried out to examine the beneficial impacts of this hybridization. Consequently, it is important to maintain strong control over the choices and parameters used in the manufacturing of polymer nanocomposites. It is also necessary to thoroughly study how NPs affect the environment to use NPs safely and prevent environmental dangers.

Research ethics: Hereby, we consciously assure that for the manuscript titled "A brief review on Polymer nanocomposite: Current trends and prospects", the following is fulfilled: (1) This material is the authors' own original work, which has not been published elsewhere. (2) The paper is not currently being considered for publication elsewhere.

Informed consent: Not applicable as this doesn't involve any clinical investigation.

Author contributions: All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by [I. Siva Ramakoti], and [Narayan Gouda]. The first draft of the manuscript was written by [I. Siva Ramakoti] and [Achyut Kumar Panda], reviewed and corrected the first draft. All authors read and approved the final manuscript.

Competing interests: The authors have no relevant financial or non-financial interests to disclose.

Research funding: The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Data availability: Not applicable as the manuscript does not include any data.

References

- Komarneni S. *J Mater Chem* 1992, 2, 1219–1230.
- Hazarika D., Karak N. Fundamentals of polymeric nanostructured materials. In *Advances in Polymeric Nanomaterials for Biomedical Applications*; Bajpai A. K., Saini R. K., Eds.; Elsevier Inc: Amsterdam, Netherlands, 2021; pp. 1–40.
- Fu S., Sun Z., Huang P., Li Y., Hu N. *Nano Mater. Sci.* 2019, 1, 2–30.
- Koo J. H. *Fundamentals, Properties, and Applications of Polymer Nanocomposites*; Cambridge University Press: Cambridge, England, 2017.
- Koo J. H. *Polymer Nanocomposites: Processing, Characterization, and Applications*, 1st ed.; McGraw-Hill Education: New York, 2006.
- Mallakpour S., Behranvand V. *Hybrid Polymer Composite Materials: Applications*; Elsevier Inc.: Amsterdam, Netherlands, 2017; pp. 197–226.
- Zhao M., Wu H. M., Zhu Z., Wu J. L., Kang W. H., Sue H. J. *Macromolecules* 2022, 55, 3039–3050.
- Jardón-Maximino N., Cadenas-Pliego G., Ávila-Orta C. A., Comparán-Padilla V. E., Lugo-Urbe L. E., Pérez-Alvarez M., Tavizón S. F., Dejesús Sosa Santillán G. *Polymers* 2021, 13, 1694.
- Jeon J. G., Kim H. C., Kim J., Kang T. J. *Funct. Compos. Struct.* 2020, 2, 015002.
- Abdel-Gawad N. M. K., El Dein A. Z., Mansour D. E. A., Ahmed H. M., Darwish M. M. F., Lehtonen M. *High Voltage* 2020, 5, 463–471.
- Zhu Z., Tsai C. Y., Zhao M., Baker J., Sue H. J. *Macromolecules* 2022, 55, 1165–1177.
- Xu H., Fan P., Xu L. *Catalysts* 2023, 13, 110.
- Shokri A. A., Talebi S., Salami-Kalajahi M. *Ind. Eng. Chem. Res.* 2020, 59, 15202–15213.
- Cai X., Jiang Z., Zhang X., Gao T., Yue K., Zhang X. *RSC Adv.* 2018, 8, 11367–11374.
- Santos B. P. S., Arias J. J. R., Jorge F. E., de Deus Santos R. E. P., da Silva Fernandez B., da Silva Candido L., de Carvalho Peres A. C., Chaves É. G., Marques M. de F. V. J. *Mater. Res. Technol.* 2021, 15, 3547–3557.
- Li Z., Fang C., Qian C., Zhou S., Song X., Ling M., Liang C., Liu G. *ACS Appl. Polym. Mater.* 2019, 1, 1965–1970.
- Roy K., Alam M. N., Mandal S. K., Debnath S. C. *Polym. Bull.* 2016, 73, 191–207.
- Taraghi I., Paszkiewicz S., Irska I., Pypec K., Piesowicz E. *Polymers* 2020, 12, 1–10.
- Castro-Landinez J. F., Salcedo-Galan F., Medina-Perilla J. A., Leporatti S., Fakhruddin R. F. *Polymers* 2021, 13, 705.
- Ibrahim M. A. H., Hassan A., Wahit M. U., Hasan M., Mokhtar M. *J. Elastomers Plast.* 2017, 49, 209–225.
- Sinclair A., Zhou X., Tangpong S., Bajwa D. S., Quadir M., Jiang L. *ACS Omega* 2019, 4, 13189–13199.
- Atanassov A., Kostov G., Kiryakova D., Borisova-Koleva L. *J. Thermoplast. Compos. Mater.* 2012, 27, 126–141.
- Ziabka M., Dziadek M., Menaszek E. *Polymers* 2018, 10, 1257.
- Allel A., Benguergoura H., Naceur M. W., Ledoux A., Saeed W. S., Aouak T. *Membranes* 2021, 11, 921.
- Karahaliou E.-K., Tarantili P. A. *J. Appl. Polym. Sci.* 2009, 113, 2271–2281.
- Youssef A. M., El-Gendy A., Kamel S. *Mater. Chem. Phys.* 2015, 152, 26–33.
- Mallakpour S., Behranvand V. *Des. Monomers Polym.* 2016, 19, 283–289.
- Mallakpour S., Javadpour M. *J. Appl. Polym. Sci.* 2016, 133, 1–9.
- Herrera-Sandoval G. M., Baez-Angarita D. B., Correa-Torres S. N., Primera-Pedrozo O. M., Hernández-Rivera S. P. *Mater. Sci. Appl.* 2013, 04, 179–185.
- Chowreddy R. R., Nord-Varhaug K., Rapp F. J. *Polym. Environ.* 2019, 27, 37–49.
- Satya S. K., Sreekanth P. S. R. *Iran. Polym. J.* 2022, 31, 629–640.
- Pastoriza-Santos I., Kinnear C., Pérez-Juste J., Mulvaney P., Liz-Marzán L. M. *Nat. Rev. Mater.* 2018, 3, 375–391.
- Ismayil K. M., Varghese A., Antony R. J. *Elastomers Plast.* 2020, 52, 103–116.

34. Awad M. A., Hendi A. A., Ortashi K. M. O., Alanazi A. B., ALZahrani B. A., Soliman D. A. *Int. J. Polym. Sci.* 2019, 2019, 1–7.
35. Prasanna S. R. V. S., Balaji K., Pandey S., Rana S. *Nanomaterials and Polymer Nanocomposites: Raw Materials to Applications*; Elsevier: Amsterdam, Netherlands, 2019; pp. 123–144.
36. Maharramov A. A., Ramazanov M. A., di Palma L., Shirinova H. A., Hajiyeve F. v. *Russ. Phys. J.* 2018, 60, 1572–1576.
37. Uddin Md. N., Desai F. J., Asmatulu E. *Energy Ecol. Environ.* 2020, 5, 1–11.
38. Lalegül-Ülker Ö., Elçin Y. M. *Mater. Sci. Eng. C* 2021, 119, 111600.
39. Kausar A. *Clay Nanoparticles*; Elsevier: Amsterdam, Netherlands, 2020; pp. 169–184.
40. Joussein E., Petit S., Churchman J., Theng B., Righi D., Delvaux B. *Clay Miner* 2005, 40, 383–426.
41. Galimberti M., Ed. *Rubber-Clay Nanocomposites*; John Wiley & Sons, Inc: Hoboken, NJ, USA, 2011.
42. Kickelbick G. *Prog. Polym. Sci.* 2003, 28, 83–114.
43. Kar B., Rout D. Nanoclays: synthesis, properties and applications. In *Polymer Nanocomposites Based on Inorganic and Organic Nanomaterials*; Mohanty S., Nayak S. K., Kaith B. S., Kalia S., Eds. John Wiley & Sons, Inc: Hoboken, NJ, USA, 2015; pp. 195–214.
44. Oliveira M., Machado A. V. *Preparation of Polymer-Based Nanocomposites by Different Routes*; Nova Science Publishers: Hauppauge, New York., 2013; pp. 1–22.
45. Suresh R., Borkar S. N., Sawant V. A., Shend V. S., Dimble S. K. *Int. J. Pharm. Sci. Nanotechnol.* 2010, 3, 901–906.
46. Hwang S. J., Joo Y. L., Lee S. J. *J. Appl. Polym. Sci.* 2008, 110, 1441–1450.
47. Halim K. A. A., Farrell J. B., Kennedy J. E. *Mater. Chem. Phys.* 2013, 143, 336–348.
48. Lei F., Yang S., Yang M., Li J., Guo S. *Polym. Bull.* 2014, 71, 3261–3273.
49. Zhang S., Horrocks A. R., Hull R., Kandola B. K. *Polym. Degrad. Stab.* 2006, 91, 719–725.
50. Lagaron J. M., Cabedo L., Cava D., Feijoo J. L., Gavara R., Gimenez E. *Food Addit. Contam.* 2005, 22, 994–998.
51. Gill Y. Q., Song M., Abid U. *Polym. Bull.* 2020, 77, 3749–3765.
52. Schindelholz E. J., Spoerke E. D., Nguyen H.-D., Grunlan J. C., Qin S., Bufford D. C. *ACS Appl. Mater. Interfaces* 2018, 10, 21799–21803.
53. Mrah L., Meghabar R. *Polym. Bull.* 2021, 78, 3509–3526.
54. Díez-Pascual A. M. *Polymers* 2020, 12, 872.
55. Abbasi H., Antunes M., Velasco J. I. *Prog. Mater. Sci.* 2019, 103, 319–373.
56. Asadian E., Ghalkhani M., Shahrokhian S. *Sens. Actuators, B Chem.* 2019, 293, 183–209.
57. Byrne M. T., Gun'ko Y. K. *Adv. Mater.* 2010, 22, 1672–1688.
58. de Volder M. F. L., Tawfick S. H., Baughman R. H., Hart A. J. *Science* 2013, 339, 535–539.
59. Sahoo N. G., Rana S., Cho J. W., Li L., Chan S. H. *Prog. Polym. Sci.* 2010, 35, 837–867.
60. Ajayan P. M., Stephan O., Colliex C., Trauth D. *Science* 1994, 265, 1212–1214.
61. Yadav R., Tirumali M., Wang X., Naebe M., Kandasubramanian B. *Def. Technol.* 2020, 16, 107–118.
62. Cheng H., Wei S., Ji Y., Zhai J., Zhang X., Chen J., Shen C. *Compos. Appl. Sci. Manuf.* 2019, 121, 139–148.
63. Zhao Y., Hou J., Bai Z., Yang Y., Guo X., Cheng H., Zhao Z., Zhang X., Chen J., Shen C. *Compos. Appl. Sci. Manuf.* 2020, 139, 106095.
64. Kashiwagi T., Du F., Winey K. I., Groth K. M., Shields J. R., Bellayer S. P., Kim H., Douglas J. F. *Polymers* 2005, 46, 471–481.
65. Gupta T. K., Singh B. P., Dhakate S. R., Singh V. N., Mathur R. B. *J. Mater. Chem. A Mater.* 2013, 1, 9138–9149.
66. Trivedi D. N., Rachchh N. v. *Polymer* 2022, 240, 124486.
67. Elastic R. J., Kinloch I. A., Gong L., Novoselov K. S. *Compos. Sci. Technol.* 2012, 72, 1459–1476.
68. Zhu Y., Murali S., Cai W., Li X., Suk J. W., Potts J. R., Ruoff R. S. *Adv. Mater.* 2010, 22, 3906–3924.
69. Gong L., Kinloch I. A., Elastic R. J., Riaz I., Jalil R., Novoselov K. S. *Adv. Mater.* 2010, 22, 2694–2697.
70. Legge E. J., Paton K. R., Wywijas M., McMahon G., Pemberton R., Kumar N., Aranga Raju A. P., Dawson C. P., Strudwick A. J., Bradley J. W., Stolojan V., Silva S. R. P., Hodge S. A., Brennan B., Pollard A. J. *ACS Appl. Mater. Interfaces* 2020, 12, 13481–13493.
71. Zhao B., Zhao C., Hamidinejad M., Wang C., Li R., Wang S., Yasamin K., Park C. B. *J. Mater. Chem. C Mater.* 2018, 6, 10292–10300.
72. Gupta S., Tai N.-H. *Carbon* 2019, 152, 159–187.
73. Balandin A. A. *Nat. Mater.* 2011, 10, 569–581.
74. Leeladhar Raturi P., Singh J. P. *Sci. Rep.* 2018, 8, 3687.
75. Yu L., Yu H. *ACS Appl. Mater. Interfaces* 2015, 7, 3834–3839.
76. Wang G., Fu Y., Guo A., Mei T., Wang J., Li J., Wang X. *Chem. Mater.* 2017, 29, 5629–5635.
77. Ma T., Liu Z., Wen J., Gao Y., Ren X., Chen H., Jin C., Ma X.-L., Xu N., Cheng H.-M., Ren W. *Nat. Commun.* 2017, 8, 1–9.
78. Shen X., Wang Z., Wu Y., Liu X., He Y.-B., Zheng Q., Yang Q.-H., Kang F., Kim J.-K. *Mater. Horiz.* 2018, 5, 275–284.
79. Wu Z., Xu C., Ma C., Liu Z., Cheng H., Ren W. *Adv. Mater.* 2019, 31, 1900199.
80. Layek R. K., Nandi A. K. *Polymer* 2013, 54, 5087–5103.
81. Park C. M., Kim Y. M., Kim K.-H., Wang D., Su C., Yoon Y. *Chemosphere* 2019, 221, 392–402.
82. Wang X., Liu X., Yuan H., Liu H., Liu C., Li T., Yan C., Yan X., Shen C., Guo Z. *Mater. Des.* 2018, 139, 372–379.
83. Tang M., Xing W., Wu J., Huang G., Li H., Wu S. *Chin. J. Polym. Sci.* 2014, 32, 658–666.
84. Dul S., Fambri L., Pegoretti A. *Compos. Appl. Sci. Manuf.* 2016, 85, 181–191.
85. Chen Y., Yao X., Zhou X., Pan Z., Gu Q. *J. Nanosci. Nanotechnol.* 2011, 11, 7813–7819.
86. Kim H., Macosko C. W. *Polymer* 2009, 50, 3797–3809.
87. Mittal V., Kim S., Neuhofer S., Paulik C. *Colloid Polym. Sci.* 2016, 294, 691–704.
88. Bafana A. P., Yan X., Wei X., Patel M., Guo Z., Wei S., Wujcik E. K. *Compos. B Eng.* 2017, 109, 101–107.
89. Koutsoumpis S., Klonos P., Raftopoulos K. N., Papadakis C. M., Bikiaris D., Pissis P. *Polymer* 2018, 153, 548–557.
90. Wilczewski S., Skórczewska K., Tomaszewska J., Lewandowski K. *Polym. Test.* 2020, 81, 106282.
91. Sagbas S., Sahiner N. *Nanocarbon and its Composites*; Elsevier: Amsterdam, Netherlands, 2019; pp. 651–676.
92. Bakirhan N. K., Ozkan S. A. *Handbook of Nanomaterials for Industrial Applications*; Elsevier: Amsterdam, Netherlands, 2018; pp. 520–529.
93. Xia C., Zhu S., Feng T., Yang M., Yang B., Xia C., Feng T., Yang M., Yang B., Zhu S. *Adv. Sci.* 2019, 6, 1901316.
94. Mansuriya B. D., Altintas Z. *Materials* 2019, 13, 96.
95. Saliha D., Özmelleş R. G. *Turk. J. Anal. Chem.* 2020, 2, 37–46.
96. Singh I., Arora R., Dhiman H., Pahwa R. *Turk. J. Pharm. Sci.* 2018, 15, 219.
97. Wang Y., Hu A. J. *Mater. Chem. C Mater.* 2014, 2, 6921–6939.
98. Sciortino A., Cannizzo A., Messina F. C. *J. Carbon Res.* 2018, 4, 67.
99. Namdari P., Negahdari B., Eatemadi A. *Biomed. Pharmacother.* 2017, 87, 209–222.
100. Tuerhong M., Yang X. U., Xue-Bo Y. I. N. *Chin. J. Anal. Chem.* 2017, 45, 139–150.
101. Zhou Y., Sharma S. K., Peng Z., Leblanc R. M. *Polymers* 2017, 9, 67.

102. Feng Z., Adolffson K. H., Xu Y., Fang H., Hakkarainen M., Wu M. *Sustain. Mater. Technol.* 2021, 29, e00304.
103. Zulfajri M., Sudewi S., Smulyati S., Rasool A., Adlim M., Huang G. G. *Coatings* 2021, 11, 1100.
104. Hu T., Wen Z., Wang C., Thomas T., Wang C., Song Q., Yang M. *Nanoscale Adv.* 2019, 1, 1413–1420.
105. Wang C., Hu T., Chen Y., Xu Y., Song Q. *ACS Appl. Mater. Interfaces* 2019, 11, 22332–22338.
106. Tian Z., Li D., Ushakova E. v., Maslov V. G., Zhou D., Jing P., Shen D., Qu S., Rogach A. L. *Adv. Sci.* 2018, 5, 1800795.
107. Li Y., Li S., Zhang K. J. *Membr. Sci.* 2017, 537, 42–53.
108. Safaei B., Youssefi M., Rezaei B., Irannejad N. *Smart Sci.* 2018, 6, 117–124.
109. Putro P. A., Yudasari N., Isnaeni I., Maddu A. *Walailak J. Sci. Technol.* 2021, 18.
110. Fernandes D., Heslop K. A., Kellarakis A., Krysmann M. J., Estevez L. *Polymer* 2020, 188, 122159.
111. Carvalho J., Santos L. R., Germino J. C., Terezo A. J., Moreto J. A., Quites F. J., Freitas R. G. *Mater. Res.* 2019, 22, 1–8; <https://doi.org/10.1590/1980-5373-mr-2018-0920>.
112. Cao G. *Nanostructures and Nanomaterials - Synthesis, Properties and Applications*; Imperial College Press: London, 2004.
113. Karak N. *Experimental Methods on Polymers, Nanomaterials and Their Nanocomposites*; Nova Publishers: Hauppauge, New York, 2016.
114. Karak N. *Nanomaterials and Polymer Nanocomposites: Raw Materials to Applications*; Elsevier: Amsterdam, Netherlands, 2018; pp. 1–45.
115. Ijaz I., Gilani E., Nazir A., Bukhari A. *Green Chem. Lett. Rev.* 2020, 13, 223–245.
116. Salah N., Khan H., Memic A., Al-Hamed S., Habib Z. *Int. J. Nanomed.* 2011, 6, 863–869.
117. Carneiro J. O., Azevedo S., Fernandes F., Freitas E., Pereira M., Tavares C., Lanceros-Méndez S., Teixeira V. J. *Mater. Sci.* 2014, 49, 7476–7488.
118. Yang B., Ding C. *Process. Appl. Ceram.* 2017, 11, 39–44.
119. Amendola V., Meneghetti M. *Phys. Chem. Chem. Phys.* 2009, 11, 3805.
120. Al-Azawi M. A., Bidin N. *Chin. J. Phys.* 2015, 53, 201–209.
121. Singh A., Vihinen J., Frankberg E., Hyvärinen L., Honkanen M., Levänen E. *Nanoscale Res. Lett.* 2016, 11, 1–9.
122. Colomer J.-F., Stephan C., Lefrant S., van Tendeloo G., Willems I., Kónya Z., Fonseca A., Laurent C., Nagy J. B. *Chem. Phys. Lett.* 2000, 317, 83–89.
123. Kasumov Y. A., Shailos A., Khodos I. I., Volkov V. T., Levashov V. I., Matveev V. N., Guéron S., Kobylko M., Kociak M., Bouchiat H., Agache V., Rollier A. S., Buchaillot L., Bonnot A. M., Kasumov A. Y. *Appl. Phys. A* 2007, 88, 687–691.
124. Cao Z., Sun Z., Guo P., Chen Y. *Front. Mater. Sci. China* 2007, 1, 92–96.
125. Magrez A., Smajda R., Seo J. W., Horváth E., Ribič P. R., Andresen J. C., Acquaviva D., Olariu A., Laurenczy G., Forró L. *ACS Nano* 2011, 5, 3428–3437.
126. Byrappa K., Adschiri T. *Prog. Cryst. Growth Char. Mater.* 2007, 53, 117–166.
127. Mascolo M., Pei Y., Ring T. *Materials*. 2013, 6, 5549–5567.
128. Petcharoen K., Sirivat A. *Mater. Sci. Eng. B: Solid-State Mater. Adv. Technol.* 2012, 177, 421–427.
129. Maaz K., Karim S., Mumtaz A., Hasanain S. K., Liu J., Duan J. L. J. *Magn. Magn. Mater.* 2009, 321, 1838–1842.
130. Pradeep A., Priyadharsini P., Chandrasekaran G. J. *Magn. Magn. Mater.* 2008, 320, 2774–2779.
131. Xu J., Yang H., Fu W., Du K., Sui Y., Chen J., Zeng Y., Li M., Zou G. J. *Magn. Magn. Mater.* 2007, 309, 307–311.
132. Daou T. J., Pourroy G., Bégin-Colin S., Grenèche J. M., Ulhaq-Bouillet C., Legaré P., Bernhardt P., Leuvrey C., Rogez G. *Chem. Mater.* 2006, 18, 4399–4404.
133. Chiu H.-C., Yeh C.-S. *J. Phys. Chem. C* 2007, 111, 7256–7259.
134. Arumugham T., Alagumuthu M., Amimodu R. G., Munusamy S., Iyer S. K. *Sustain. Mater. Technol.* 2020, 23, e00138.
135. Kasibabu B. S. B., D'souza S. L., Jha S., Kailasa S. K. J. *Fluoresc.* 2015, 25, 803–810.
136. D'souza S. L., Chettiar S. S., Koduru J. R., Kailasa S. K. *Optik* 2018, 158, 893–900.
137. Ramar V., Moothattu S., Balasubramanian K. *Sol. Energy* 2018, 169, 120–127.
138. Rane A. V., Kanny K., Abitha V. K., Thomas S. *Synthesis of Inorganic Nanomaterials*; Elsevier: Amsterdam, Netherlands, 2018; pp. 121–139.
139. Rong M. Z., Zhang M. Q., Zheng Y. X., Zeng H. M., Friedrich K. *Polymer* 2001, 42, 3301–3304.
140. Liu X., Wu Q. *Polymer* 2001, 42, 10013–10019.
141. Safdari M., Al-Haik M. S. *Carbon-Based Polymer Nanocomposites for Environmental and Energy Applications*; Elsevier: Amsterdam, Netherlands, 2018; pp. 113–146.
142. Das R., Pattanayak A. J., Swain S. K. *Polymer-based Nanocomposites for Energy and Environmental Applications*; Elsevier: Amsterdam, Netherlands, 2018; pp. 205–218.
143. Rane A. V., Kanny K., Abitha V. K., Thomas S. *Synthesis of Inorganic Nanomaterials*; Elsevier: Amsterdam, Netherlands, 2018; pp. 121–139.
144. Cheng S., Grest G. S. *ACS Macro Lett.* 2016, 5, 694–698.
145. Bivens A. *Polymer-to-Solvent Reference Table for GPC/SEC Technical Overview Author*. 5991-6802EN.indd (agilent.com) (accessed June 15, 2023).
146. Anandhan S., Bandyopadhyay S. *Nanocomposites and Polymers with Analytical Methods*; InTech: London, 2011; pp. 3–28.
147. Vaia R. A., Giannelis E. P. *Macromolecules* 1997, 30, 8000–8009.
148. Moniruzzaman M., Winey K. I. *Macromolecules* 2006, 39, 5194–5205.
149. Grossiord N., Kivit P. J. J., Loos J., Meuldijk J., Kyrlyuk A. v., van der Schoot P., Koning C. E. *Polymer* 2008, 49, 2866–2872.
150. Potts J. R., Dreyer D. R., Bielawski C. W., Ruoff R. S. *Polymer* 2011, 52, 5–25.
151. Naz A., Kausar A., Siddiq M., Choudhary M. A. *Polym. Plast. Technol. Eng.* 2016, 55, 171–198.
152. Mallakpour S., Khadem E. *Prog. Polym. Sci.* 2015, 51, 74–93.
153. Spitalsky Z., Tasis D., Papagelis K., Galiotis C. *Prog. Polym. Sci.* 2010, 35, 357–401.
154. il Jo C., Ko J., Yin Z., Kim Y.-J., Kim Y. S. *Ind. Eng. Chem. Res.* 2016, 55, 9433–9439.
155. Qi D., Liu C., Chen Z., Dong G., Cao Z. *Colloid Polym. Sci.* 2015, 293, 463–471.
156. Mallakpour S., Naghdi M. *Prog. Mater. Sci.* 2018, 97; <https://doi.org/10.1016/j.pmatsci.2018.04.002>.
157. Fornes T. D., Yoon P. J., Keskkula H., Paul D. R. *Polymer* 2001, 42, 9929–9940.
158. la Mantia F. P., Mistretta M. C., Morreale M. *Macromol. Mater. Eng.* 2014, 299, 96–103.
159. de Oliveira A. D., Beatrice C. A. G. *Nanocomposites – Recent Evolutions*; Intech Open: London, 2019; pp. 104–128.
160. Kickelbick G. *Hybrid Mater.* 2007, 1, 2.
161. Oliveira M., Machado A. v. *Nanocomposites: Synthesis, Characterization and Applications*; Nova Science Publishers, Inc.: Hauppauge, New York, 2013; pp. 73–94.
162. Konwarh R., Karak N., Misra M. *Biotechnol. Adv.* 2013, 31, 421–437.
163. Khan I., Saeed K., Khan I. *Arab. J. Chem.* 2019, 12, 908–931.

164. Townsend E., Bryant G. W. *Nano Lett.* 2012, 12, 429–434.
165. Raveendran P., Fu J., Wallen S. L. *J. Am. Chem. Soc.* 2003, 125, 13940–13941.
166. Faupel F., Zaporojtchenko V., Strunskus T., Elbahri M. *Adv. Eng. Mater.* 2010, 12, 1177–1190.
167. Abbasi N. M., Yu H., Wang L., Zain-ul-abdin Amer W. A., Sun R., Akram M., Khalid H., Chen Y., Saleem M., Sun R., Shan J. *Mater. Chem. Phys.* 2015, 166, 1–15.
168. Wankhede Y. B., Kondawar S. B., Thakare S. R., More P. S. *Adv. Mater. Lett.* 2013, 4, 89–93.
169. Huang X., Jiang P., Xie L. *Appl. Phys. Lett.* 2009, 95, 242901.
170. Li D., He Q., Li J. *Adv. Colloid. Interface. Sci.* 2009, 149, 28–38.
171. O'Neal J. T., Bolen M. J., Dai E. Y., Lutkenhaus J. L. *J. Colloid Interface Sci.* 2017, 485, 260–268.
172. Burunkova J., Denisiuk I., Vorzobova N., Daroczi L., Hegedus C., Charnovych S., Kokenyesi S. *Eur. Polym. J.* 2013, 49, 3072–3077.
173. Zhang R. Y., Olin H. *Int. J. Biomed. Nanosci. Nanotechnol.* 2011, 2, 112.
174. Mikoliūnaitė L., Kubiliūtė R., Popov A., Voronovič J., Šakirzanovas S., Ramanavičienė A., Ramanavičius A., Mikoliūnaitė L., Kubiliūtė R., Popov A., Voronovič J., Šakirzanovas S., Ramanavičienė A., Ramanavičius A. *Chemija* 2014, 25, 63–69.
175. Abdelrasoul G. N., Farkas B., Romano I., Diaspro A., Beke S. *Mater. Sci. Eng. C* 2015, 56, 305–310.
176. Husain J., Reddy N., Sagar J. *Int. J. Eng. Sci. Technol.* 2016, 2, 58–63.
177. Sharma H., Jain S., Raj P. M., Murali K. P., Tummala R. J. *Electron. Mater.* 2015, 44, 3819–3826.
178. Bogdanović U., Vodnik V., Mitrić M., Dimitrijević S., Škapin S. D., Žunić V., Budimir M., Stojiljković M. *ACS Appl. Mater. Interfaces* 2015, 7, 1955–1966.
179. Kuo W. S., Chang C. N., Chang Y. T., Yang M. H., Chien Y. H., Chen S. J., Yeh C. S. *Angew. Chem. Int. Ed.* 2010, 49, 2711–2715.
180. Opoku F., Kiarri E. M., Govender P. P., Mamo M. A. *Descriptive Inorganic Chemistry Researches of Metal Compounds*; InTech: London, 2017; pp. 173–199.
181. Rane A. V., Kanny K., Abitha V. K., Thomas S., Thomas S. *Synthesis of Inorganic Nanomaterials: Advances and Key Technologies*; Elsevier Inc.: Amsterdam, Netherlands, 2018; pp. 121–139.
182. Lu X., Mao H., Chao D., Zhang W., Wei Y. J. *Solid State Chem.* 2006, 179, 2609–2615.
183. Wu T., Ke Y. *Thin Solid Films* 2007, 515, 5220–5226.
184. Hashimoto M., Takadama H., Mizuno M., Kokubo T. *Mater. Res. Bull.* 2006, 41, 515–524.
185. Wu Y., Song L., Hu Y. *Ind. Eng. Chem. Res.* 2011, 50, 11988–11995.
186. Mallakpour S., Javadpour M. *Polym. Adv. Technol.* 2017, 28, 393–403.
187. Sengwa R. J., Dhatarwal P. *Opt. Mater.* 2021, 113, 110837.
188. Al-Bataineh Q. M., Migdadi A. B., Telfah A., Ahmad A. A., Alsaad A. M., Tavares C. J. *Mater. Chem. Phys.* 2022, 290, 126387.
189. Julkapli N. M., Bagheri S. *Int. J. Hydrogen Energy* 2015, 40, 948–979.
190. In J. B., Noy A. *Nanotechnology's Wonder Material: Synthesis of Carbon Nanotubes*. In *Hierarchical Nanostructures for Energy Devices*; Ko S. H., Grigoropoulos C. P., Eds.; Royal Society of Chemistry: Cambridge, UK, 2014; pp. 26–58.
191. Bal S., Samal S. S. *Bull. Mater. Sci.* 2007, 30, 379–386.
192. Hu Z., Shao Q., Xu X., Zhang D., Huang Y. *Compos. Sci. Technol.* 2017, 142, 294–301.
193. Xie X. L., Mai Y. W., Zhou X. P. *Bull. Mater. Sci.* 2005, 49, 89–112.
194. Yang W., Xing W., Yang W., Hu Q., Si J., Lu H., Yang B., Song L., Hu Y., Yuen R. K. K. *ACS Appl. Mater. Interfaces* 2016, 8, 26266–26274.
195. King J. A., Klimek D. R., Miskioğlu I., Odegard G. M. *J. Appl. Polym. Sci.* 2013, 128, 4217–4223.
196. Zhu S., Song Y., Zhao X., Shao J., Zhang J., Yang B. *Nano Res.* 2015, 8, 355–381.
197. Mansuriya B. D., Altintas Z. *Nanomaterials*. 2018, 11.
198. Jeon H. R., Park J. H., Shon M. Y. *J. Ind. Eng. Chem.* 2013, 19, 849–853.
199. Liu D., Zhao W., Liu S., Cen Q., Xue Q. *Surf. Coat. Technol.* 2016, 286, 354–364.
200. Dhanaraj A., Das K., Keller J. M. *Proceedings of the International Conference on Physics and Chemistry of Materials in Novel Engineering Applications: PCMNEA 2020, KCT, Coimbatore, Ind, Feb 2020*, p. 2270.
201. Xavier M. M., Adarsh N. N., Nair P. R., Mathew S. *ACS Omega* 2021, 6, 22840–22847.
202. Eskalen H., Yaykaşlı H., Kavgacı M., Kayış A. J. *Mater. Sci. Mater. Electron.* 2022, 33, 4643–4658.
203. Zhang H., Zhu H., Xu C., Li Y., Liu Q., Wang S., Yan S. *Polymer* 2022, 252, 124944.
204. Youssef A. M. *Polym. Plast. Technol. Eng.* 2013, 52, 635–660.
205. Mentasti L., Zucchi I. A., Cammarata A., Glisoni R., Santiago M., Barreto G. *Opt. Mater.* 2022, 129, 112566.
206. Bokobza L. *Polymers* 2018, 10, 7.
207. Fu S. Y., Feng X. Q., Lauke B., Mai Y. W. *Compos. B Eng.* 2008, 39, 933–961.
208. Bazgari D., Moztarzadeh F., Sabbagh-Alvani A. A., Rasoulboroujeni M., Tahriri M., Tayebi L. *Ceram. Int.* 2018, 44, 1220–1224.
209. Machrafi H., Lebon G., Iorio C. S. *Compos. Sci. Technol.* 2016, 130, 78–87.
210. Sharifzadeh E., Tohfegar E., Safajou Jahankhanemlou M. *Iran. J. Chem. Sher. Eng.* 2020, 17, 65–78.
211. Ashraf M. A., Peng W., Zare Y., Rhee K. Y. *Nanoscale Res. Lett.* 2018, 13, 214.
212. Ansari R., Hassanzadeh-Aghdam M. K., Darvizeh A. *Mech. Mater.* 2016, 101, 14–26.
213. Msekh M. A., Cuong N. H., Zi G., Areias P., Zhuang X., Rabczuk T. *Eng. Fract. Mech.* 2018, 188, 287–299.
214. Sharifzadeh E., Ghasemi I., Karrabi M., Azizi H. *Iran. Polym. J.* 2014, 23, 835–845.
215. Sharifzadeh E., Cheraghi K. *Mech. Mater.* 2021, 160, 103990.
216. Doniavi A., Babazadeh S., Azdast T., Hasanzadeh R. J. *Elastomers Plast.* 2016, 49, 498–512.
217. Abedini A., Asiyabi T., Campbell H. R., Hasanzadeh R., Azdast T. *Int. J. Adv. Manuf. Technol.* 2019, 102, 1747–1758.
218. Ürk D., Demir E., Bulut O., Çakıroğlu D., Cebeci F., Lütfi Öveçoğlu M., Cebeci H. *Compos. Struct.* 2016, 155, 255–262.
219. Mokhireva K. A., Svistkov A. L., Solod'ko V. N., Komar L. A., Stöckelhuber K. W. *Polym. Test.* 2017, 59, 46–54.
220. Nejad S. A., Majzoobi G. H., Sabet S. A. R. *Iran. Polym. J.* 2019, 28, 895–908.
221. Gilman J. W. *Appl. Clay Sci.* 1999, 15, 31–49.
222. Bai G., Guo C., Li L. *Constr. Build. Mater.* 2014, 50, 148–153.
223. Yuan B., Sun Y., Chen X., Shi Y., Dai H., He S. *Compos. Appl. Sci. Manuf.* 2018, 109, 345–354.
224. Idris A., Muntean A., Mesic B. J. *Coat. Technol. Res.* 2022, 19, 699–716.
225. Yazman Ş., Samancı A. *Arab. J. Sci. Eng.* 2019, 44, 1353–1363.
226. Mahmoudian M., Marjani A. P., Hasanzadeh R., Moradian M., Shishavan S. M. *Polym. Bull.* 2020, 77, 2837–2854.
227. Ma L., Wang G., Dai J. *High Perform. Polym.* 2016, 29, 187–196.
228. Shen W., Zhang T., Ge Y., Feng L., Feng H., Li P. *Prog. Org. Coating* 2021, 152, 106130.
229. Suresh S., Nisha P., Saravanan P., Jayamoorthy K., Karthikeyan S. *Silicon* 2018, 10, 1291–1303.
230. Chen J., Wei H., Bao H., Jiang P., Huang X. *ACS Appl. Mater. Interfaces* 2019, 11, 31402–31410.

231. Dhatarwal P., Sengwa R. J. *Optik* 2021, 233, 166594.
232. Alsaad A. M., Ahmad A. A., Dairy A. R., al Al-anbar A. S., Al-Bataineh Q. M. *Results Phys.* 2020, 19, 103463.
233. Rasul M. G., Kiziltas A., Malliakas C. D., Rojaee R., Sharifi-Asl S., Foroozan T., Shahbazian-Yassar R., Arfaei B. *Compos. Sci. Technol.* 2021, 204, 108631.
234. Madaeni S. S., Zinadini S., Vatanpour V. *Sep. Purif. Technol.* 2013, 111, 98–107.
235. Rahimpour A., Jahanshahi M., Rajaeian B., Rahimnejad M. *Desalination* 2011, 278, 343–353.
236. Etemadi H., Afsharkia S., Zinatloo-Ajabshir S., Shokri E. *Polym. Eng. Sci.* 2021, 61, 2364–2375.
237. Vatanpour V., Khadem S. S. M., Dehqan A., Al-Naqshabandi M. A., Ganjali M. R., Hassani S. S., Rashid M. R., Saeb M. R., Dizge N. *Chemosphere* 2021, 263.
238. Huang X., Wang R., Jiao T., Zou G., Zhan F., Yin J., Zhang L., Zhou J., Peng Q. *ACS Omega* 2019, 4, 1897–1906.
239. Shawky H. A., Chae S. R., Lin S., Wiesner M. R. *Desalination* 2011, 272, 46–50.
240. Song X., Zhou Q., Zhang T., Xu H., Wang Z. *J. Mater. Chem. A Mater.* 2016, 4, 16896–16905.
241. Zinadini S., Zinatizadeh A. A., Rahimi M., Vatanpour V., Zangeneh H. *J. Memb. Sci.* 2014, 453, 292–301.
242. Sepúlveda F. A., Rivera F., Loyo C., Canales D., Moreno-Serna V., Benavente R., Rivas L. M., Ulloa M. T., Gil-Castell O., Ribes-Greus A., Ortiz J. A., Zapata P. A. *J. Appl. Polym. Sci.* 2022, 139, 51542.
243. Laourari I., Lakhdari N., Belgherbi O., Medjili C., Berkani M., Vasseghian Y., Golzadeh N., Lakhdari D. *Chemosphere* 2022, 291, 132696.
244. Prashanth K. S., Revathi V. *Mater. Today Proc.* 2021, 49, 593–596.
245. Ahmadi Y., Yadav M., Ahmad S. *Compos. B Eng.* 2019, 164, 683–692.
246. Bacali C., Baldea I., Moldovan M., Carpa R., Olteanu D. E., Filip G. A., Nastase V., Lascu L., Badea M., Constantiniuc M., Badea F. *Clin. Oral Investig.* 2020, 24, 2713–2725.
247. Liu C., Shen J., Liao C. Z., Yeung K. W. K., Tjong S. C. *Express Polym. Lett.* 2018, 12, 365–382.
248. Naskar A., Khan H., Sarkar R., Kumar S., Halder D., Jana S. *Mater. Sci. Eng. C* 2018, 91, 743–753.
249. Reishofer D., Rath T., Ehmann H. M., Gspan C., Dunst S., Amenitsch H., Plank H., Alonso B., Belamie E., Trimmel G., Spirk S. *ACS Sustain. Chem. Eng.* 2017, 5, 3115–3122.
250. Akermi M., Jaballah N., Alarifi I. M., Rahimi-Gorji M., Chaabane R., ben Ouadaben H., ben Majdoub M. J. *Mol. Liq.* 2019, 287, 110963.
251. Soliman T. S., Vshivkov S. A., Elkalashy S. I. *Opt. Mater.* 2020, 107, 110037.
252. Dhatarwal P., Sengwa R. J. *Phys. B Condens. Matter.* 2021, 613, 412989.
253. Badawi A., Alharthi S. S., Mostafa N. Y., Althobaiti M. G., Altalhi T. *Appl. Phys. A* 2019, 125, 1–9.
254. Abdelhamied M. M., Atta A., Abdelreheem A. M., Farag A. T. M., el Okr M. M. *J. Mater. Sci. Mater. Electron.* 2020, 31, 22629–22641.
255. Ebnalwaleed A. A., Sadek A. H., Ismail S. H., Mohamed G. G. *Opt. Quantum Electron.* 2022, 54, 1–31.
256. Jebur Q. M., Hashim A., Habeeb M. A. *Trans. Electr. Electron. Mater.* 2019, 20, 334–343.
257. Maity D., Kumar R. T. R. *ACS Sens.* 2018, 3, 1822–1830.
258. Liu B., Liu X., Yuan Z., Jiang Y., Su Y., Ma J., Tai H. *Sens. Actuators, B* 2019, 295, 86–92.
259. Lo M., Seydou M., Bensghaier A., Pires R., Ngingue-Sall D., Aaron J. J., Mekhalif Z., Delhalle J., Chehimi M. M. *Sensors* 2020, 20, 580.
260. Alruwais R. S., Adeosun W. A., Alsafrani A. E., Marwani H. M., Asiri A. M., Khan I., Jawaid M., Khan A. *Membranes* 2021, 11, 853.
261. Sankar S., George A., Ramesan M. T. *RSC Adv.* 2022, 12, 17637–17644.
262. Amruth K., Abhirami K. M., Sankar S., Ramesan M. T. *Inorg. Chem. Commun.* 2022, 136, 109184.
263. Pande S., Singh B. P., Mathur R. B., Dhami T. L., Saini P., Dhawan S. K. *Nanoscale Res. Lett.* 2009, 4, 327–334.
264. Al-Ghamdi A. A., Al-Hartomy O. A., Al-Salamy F., Al-Ghamdi A. A., El-Mossalamy E. H., Abdel Daiem A. M., El-Tantawy F. *J. Appl. Polym. Sci.* 2012, 125, 2604–2613.
265. Al-Saleh M. H., Sundararaj U. J. *Phys. D Appl. Phys.* 2012, 46, 035304.
266. Sit S., Nath K., Das N. C., Chakraborty G. *J. Elastomers Plast.* 2022, 54, 975–999.
267. Fallah R., Hosseinabadi S., Pourtaghi G. J. *Environ. Health Sci. Eng.* 2022, 20, 113–122.
268. Praveen M., Karthikeya G. S., Krishna R. H., Mamatha G. M., Manjunatha C., Khosla A., Nagabhushana B. M. *Diam. Relat. Mater.* 2022, 130, 109501.